Making Model Runs Auditable by Default: A Design Science Study of Signed, Replayable Capability Receipts for Organizational AI Accountability

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Abstract

Organizations remain responsible for reviewing and justifying AI-assisted business decisions, yet a retained answer alone may not establish the integrity of the evidence behind it. Motivated by accountability in refund disputes, this design-science study develops five requirements and four principles for verifiable run-level evidence and instantiates them in Lattice, an open-source runtime. Signed results can carry receipts that bind outputs and selected execution information; optional commitments cover tasks, policies, and serialized provider requests. Historical controlled evaluation tested tamper detection, retained-output verification, policy checks, and signing overhead; separate hosted cohorts examine execution and answer quality, including retrospective review of four actual public refund-dispute records. These evaluations do not establish improved managerial decisions or reduced dispute-resolution costs. Verification checks recorded commitments, not provider honesty, decision correctness, fairness, or human authorization. Stored-output verification is not model re-execution, and request commitments do not prove remote consumption. The contribution is a design for producing evidence during execution, with explicit limits on what that evidence supports. Its usefulness to managers and auditors remains to be evaluated.

Keywords: Organizational AI accountability; AI-assisted decision-making; Design science research; Audit evidence; Human oversight; AI governance

1 Introduction

Consider an illustrative refund dispute. A customer challenges a rejection recommended by an artificial intelligence (AI) system and authorized by an employee. The customer-service manager responsible for resolving the complaint must establish which policy applied, what information the system received, what it

recommended, and why the employee accepted that recommendation. If the retained record contains only the final answer, the manager cannot reconstruct those steps from that record alone. The organizational problem is how to preserve evidence that lets responsible people investigate and justify a decision. This example motivates the study; it is not a report of a deployed refund process.

In such a workflow, one model could interpret a document, another could recommend an outcome, and a tool could retrieve policy information before an employee approves or escalates the case. These contributions distribute the work, while the organization still needs an accountable decision owner and a review process. Organizational AI governance connects technical processes with organizational objectives and responsibilities (Mäntymäki et al., 2022). For the post-AI enterprise, the resulting challenge concerns human oversight and contestability: a manager must be able to question an AI-assisted outcome against the organization's rules, rather than treat the availability of an answer as sufficient justification.

Meeting that need requires more than collecting outputs. A debugging trace may record a request and response without binding them to retained inputs, the selected provider, or the policy configuration. If its integrity is not protected, a reviewer also lacks evidence that the record remained unchanged. Secure and tamper-evident logging already address important parts of this problem (Crosby and Wallach, 2009). The design gap examined here is how to apply such protections to the evidence associated with an individual model run, while making the scope of each commitment clear. Auditing research likewise treats documentation as part of an organizational process, not a substitute for one (Raji et al., 2020).

Governance frameworks reinforce the need for evidence. Article 12 of the European Union Artificial Intelligence Act addresses automatic event recording for high-risk AI systems (European Parliament and Council of the European Union, 2024). The voluntary NIST AI Risk Management Framework connects documentation and monitoring with organizational risk management (Tabassi, 2023). These frameworks motivate the requirements examined below; they do not establish that the illustrative refund workflow is legally high-risk or that a signed record demonstrates compliance. Organizations must still determine which obligations apply and how records support their review procedures.

This study addresses the evidence problem through design science research (DSR) (Hevner et al., 2004). The research question is: *How can AI-enabled business processes produce verifiable run-level evidence that supports organizational review and accountability at acceptable operational cost?* We derive requirements from governance and auditing literature, formulate design principles, and instantiate them in Lattice, an open-source runtime (Turlapati, 2026). Its central construct is the *capability receipt*: a signed record of a run's selected execution information and content commitments. The design allows reviewers to check the integrity of retained evidence offline. Producing evidence useful to managers is a design objective whose achievement requires evaluation with those users.

The study makes three contributions: requirements and principles for run-level evidence, an implementation evaluated for tamper detection, recorded-output binding, policy checks and overhead, and lessons about the limits of that evidence. Following the distinction between an artifact and its knowledge contribution (Gregor and Hevner, 2013), the transferable contribution concerns which claims a record can support and which require additional controls. Controlled and hosted evaluations supply technical evidence; they do not demonstrate improved managerial decisions or faster resolution of customer disputes.

A valid receipt establishes integrity of the signed commitments. It does not establish that an issuing runtime or provider acted honestly, that a decision was correct or fair, or that an employee authorized it. Those questions require additional evidence and organizational judgment. Verification of retained outputs is also distinct from re-executing a model. These boundaries matter when deciding how much reliance to place on a record during review.

Section 2 reviews related work; Section 3 explains the research approach. Sections 4, 5 and 6 present requirements, the public-record refund review and the artifact. Section 7 evaluates it, Section 8 discusses implications and limitations, and Section 9 concludes.

2 Background and Related Work

2.1 AI accountability, governance, and auditing

Accountability in AI denotes the obligation of an actor to explain and justify the conduct of a system to a forum that can pass judgment and impose consequences (Wieringa, 2020; Novelli et al., 2023). IS research frames AI management as a matter of autonomy, learning, and inscrutability (Berente et al., 2021), and it describes how organizations envelop AI with sociotechnical controls to contain its behavior (Asatiani et al., 2021). Organizational AI governance comprises the structures, processes, and procedures that align AI use with organizational and regulatory goals (Mäntymäki et al., 2022; Schneider et al., 2022; Papagiannidis et al., 2025). Across these accounts, governance depends on evidence. Policies and controls mean little unless an organization can show, for specific decisions, that the controls operated.

The AI auditing literature makes this dependence concrete. Raji et al. (2020) propose an end-to-end internal audit framework and stress documentation artifacts throughout the development life cycle. Mökander and Floridi (2021) argue for ethics-based auditing as a governance mechanism, and Mökander et al. (2023) propose a three-layered approach to auditing large language models at the governance, model, and application layers. Brundage et al. (2020) call for mechanisms, including audit trails, that make claims about AI development verifiable by outsiders. Transparency research cautions that transparency must be designed for specific audiences and purposes (Felzmann et al., 2019; Rai, 2020), and human-centered AI calls for reliable, safe, and trustworthy systems backed by audit trails (Shneiderman, 2020). Documentation instruments such as model cards (Mitchell et al., 2019) and datasheets for datasets (Gebru et al., 2021) standardize disclosure about models and data. However, they describe artifacts at release time, not individual production runs. The application layer of Mökander et al. (2023), where a model is used in a specific business process, is where the evidence gap is widest, and it is the layer this study targets.

2.2 Tamper-evident logging, provenance, and transparency

Computer security offers a long tradition of tamper-evident records. Haber and Stornetta (1991) showed how to time-stamp digital documents, and Merkle (1988) introduced hash-tree commitments. Schneier and Kelsey (1999) designed secure audit logs whose entries cannot be undetectably altered after compromise, and Crosby and Wallach (2009) gave efficient data structures for tamper-evident logs with logarithmic proofs. Certificate Transparency (Laurie et al., 2013) publishes certificates to append-only logs so that misissuance

becomes publicly detectable. Software supply chain research applies these ideas to builds. in-toto (Torres-Arias et al., 2019) signs each step of a software supply chain, The Update Framework addresses survivable key compromise (Samuel et al., 2010), Sigstore makes signing and transparency logging accessible to developers (Newman et al., 2022), and SLSA defines levels of supply chain integrity (OpenSSF, 2023). The Dead Simple Signing Envelope (DSSE) specifies a minimal signed envelope with a pre-authentication encoding (Secure Systems Lab, 2021). Data provenance research models the derivation history of data products (Moreau et al., 2011; Herschel et al., 2017).

These lines of work supply mechanisms (canonical encodings, signatures, hash chains, transparency logs) but were not designed for the semantics of a model run. A model run has a declared intent, a routing decision among providers, policy constraints, output invariants, and multimodal inputs that may contain personal data. Model gateways and observability libraries such as LiteLLM (BerriAI, 2023), LangChain callbacks (LangChain Inc., 2023), OpenLLMetry (Traceloop, 2023), and the Portkey gateway (Portkey AI, 2024) sit on that path and log the calls, but none of the four signs its log, and the logs that could be read stored the prompt text. Section 7.8 reports these logs and the wrappers' latency on a CPU-only host, where generation noise dominates.

2.3 Verifiable inference and model routing

A complementary stream seeks cryptographic guarantees that a specific model computed a specific output, using zero-knowledge proofs (Kang et al., 2022; Sun et al., 2024). Such proofs give strong guarantees but currently require access to model weights and impose high costs, so they do not apply to proprietary hosted models consumed through application programming interfaces (APIs). Work on model routing optimizes cost and quality by selecting among models (Chen et al., 2023; Ong et al., 2024), often with learned routers whose decisions are hard to audit. Security research on LLM-integrated applications shows that model outputs can be manipulated through indirect prompt injection (Greshake et al., 2023), which raises the stakes for run-level evidence.

2.4 Agent receipts and verifiable agent provenance

Three recent proposals are closest to this study. *TrustTrack* is a vision paper that argues for verifiable identity, policy commitments, and tamper-resistant behavioral logs in multi-agent systems (Li, 2025). *Sello* ("Notarized Agents") inverts the trust boundary: the service that receives an agent's call signs a receipt, encrypts it to the agent's owner, and publishes it to a witness-cosigned Merkle log, so that the owner need not trust the agent or its operator (Figuera, 2026). *Janus* places a signed, hash-chained log on the effect path of agentic LLMs, so that a step's proposal and verdict are durable before its effect is released. Its gates are pure functions of the log and can be re-derived offline, and it is evaluated under crash injection and with a lending workflow (Arslan, 2026). Two further systems are run in the comparison of Section 7.7 but are not tabulated here: Agent Flight Recorder, which keeps tamper-evident audit trails of tool-using agents with on-chain anchoring (Bindschaedler et al., 2026), and NovaFabric, which seals tamper-evident, replayable evidence for autonomous agent runs (Seyedkazemi Ardebili, 2026).

Table 1: Comparison of run-level and agent-level accountability approaches (based on each system's published description; Section 7.7 reports what was measured)

Feature	TrustTrack	Sello	Janus	Lattice (this study)
Unit of record	Agent behavior	Agent action at receiving service	Agent step (proposal, verdict, effect)	Model run or terminal agent result; optional task, policy, and request bindings
Who signs	Protocol participants (design)	Receiving service	Agent runtime, validators, humans	Issuing runtime (self-attested)
Tamper evidence	Proposed	Signature and witness-cosigned Merkle log	Signed hash chain	Signature over canonical JSON; parent links; crew chains list their members
Third-party transparency log	Not specified	Yes	No	No
Deterministic, recorded routing across providers	No	No	No	Yes; selected route signed, policy hash supported, candidates and rationale outside receipt
Redaction before signing	Not specified	Encryption to owner	Not a focus	Yes, masked before signing (Sect. 7.5)
Signed output binding	Not specified	Signed receipt over the action	Signed hash chain; verdicts re-derived offline	Signed output hash, checked by re-hashing the kept output
Effect gating	No	No	Yes	No; output invariants run after generation and fail the run
Implementation	None (blueprint)	Reference implementation	Implementation	Open-source library and command-line tool
Evaluation	Use cases	Cryptographic microbenchmarks	Crash injection, 100M-event verification, model experiment	Tamper suite, output binding, policy checks, overhead (Sect. 7)

Table 1 compares these systems with the artifact presented here. The contribution is a combination of features aimed at *organizational* model use: issuer-side receipts for signed calls, terminal single-agent results, and crews; deterministic routing with a signed selected route; redaction before signing; and content commitments. Lattice still lacks Sello's receiver-side attestation and transparency log and Janus's effect gating and crash-consistency evaluation. Its current optional task, policy, and request bindings remain issuer claims (Section 6). Lattice was designed independently. Receipt code first appears in the project repository (Turlapati, 2026) in commit `fa6d5a1` of 11 May 2026, before the Sello (June 2026) and Janus (September 2026) preprints appeared. Audience, nonce, and expiry binding, chain verification, chain closure, and the later execution-binding extensions were added in October 2026, after both.

Table 2: Mapping of DSRM activities (Peffers et al., 2007) to this study

DSRM activity	This study
1. Problem identification	Accountability gap between governance expectations and model-run records (Sect. 1, 2)
2. Objectives of a solution	Design requirements DR1–DR5 (Sect. 4)
3. Design and development	Design principles DP1–DP4 instantiated in Lattice (Sect. 6)
4. Demonstration	Retrospective review of four actual public refund disputes (Sect. 5); runnable controlled scenarios: a support work inbox whose message, call transcript, policy, and photo inputs are text fixtures; a single-agent loop; and a multi-agent crew
5. Evaluation	Controlled and hosted evaluation against DR1–DR5; public-record review; planned practitioner study (Sect. 7)
6. Communication	This article; open-source artifact and evaluation harness

3 Research Approach

We follow the design science research methodology (DSRM) of Peffers et al. (2007), guided by the DSR guidelines of Hevner et al. (2004) and the contribution framework of Gregor and Hevner (2013). Table 2 maps the six DSRM activities to this study. The entry point is problem-centered: the accountability gap arose while building a browser-automation product whose autonomous actions required an auditable record, and it generalizes to any organization that embeds models in business processes. DSR builds and evaluates artifacts to solve organizational problems while producing prescriptive knowledge (vom Brocke et al., 2020), and it has long been linked to information systems design theories (Walls et al., 1992). We use the anatomy of a design theory (Gregor and Jones, 2007) to structure the knowledge contribution, with requirements as meta-requirements, principles as principles of form and function, and Lattice as an expository instantiation. Evaluation follows the Framework for Evaluation in Design Science (FEDS) of Venable et al. (2016). We adopt a technical risk and efficacy strategy, starting with artificial, summative evaluation of the instantiation against its requirements, and plan a naturalistic evaluation with practitioners as the next episode (Section 7.12). Evaluation criteria are drawn from the taxonomy of Prat et al. (2015) (goal attainment, efficiency, robustness, and consistency) and from the build-evaluate pattern of Sonnenberg and vom Brocke (2012).

4 Design Requirements and Principles

4.1 Design requirements

We derive five design requirements (DRs) from the governance, auditing, and secure logging literature. Each is stated at the level of the class of systems rather than of any particular implementation.

DR1 (Integrity and attribution). The record of each model run must be tamper-evident and attributable to an accountable issuer, so that any alteration is detectable by a party holding only public verification material. This follows from logging duties for high-risk systems (European Parliament and Council of the European Union, 2024), from the verifiable-claims agenda (Brundage et al., 2020), and from secure audit log research (Schneier and Kelsey, 1999; Crosby and Wallach, 2009).

DR2 (Reproducibility). An auditor must be able to check offline, without access to the original system

or provider, that the stored inputs and outputs of a run are the ones the record commits to. This reflects the role of reproducibility in audits (Raji et al., 2020) and in ML practice more broadly (Kapoor and Narayanan, 2023).

DR3 (Inspectable decisions and enforced policy). The record must state which provider and model served the run, and enough of the request, the policy, and the provider catalog must be kept for an auditor to recompute that choice. Declared policies (for example, privacy class, provider restrictions, budget, or citation and personal-data constraints) must be enforced, with violations recorded rather than silently retried. This reflects the inscrutability concern in AI management (Berente et al., 2021), the need for explanations tailored to auditors (Felzmann et al., 2019), and risk-management functions in Tabassi (2023).

DR4 (Data minimization). The record must not disclose secrets or personal data beyond what accountability requires, so that it can be shared with auditors without creating a new privacy risk. This reflects data protection obligations and the transparency–privacy tension discussed by Felzmann et al. (2019).

DR5 (Low adoption cost). Producing the record must add negligible runtime overhead and require little change to existing application code, because accountability mechanisms that impose high costs are bypassed in practice (Sculley et al., 2015; Amershi et al., 2019).

4.2 Design principles

We formulate four design principles (DPs), following the convention that a principle names the action, the mechanism, and the rationale (Gregor and Hevner, 2013).

DP1 (Receipt by default). Provide the runtime with a signing key so that terminal run outcomes, whether successful or failed, carry a receipt that is canonicalized, signed, and verifiable with a public key, because accountability evidence that depends on developer discipline is incomplete. (Addresses DR1, DR5.) Lattice supports this for single calls (`ai.run`), public single-agent execution (`ai.runAgent`), and multi-agent crews (`runAgentCrew`), subject to the explicit signer-failure policy (Table 4).

DP2 (Commit to content, carry hashes). Have the receipt commit to content-addressed hashes of input artifacts and outputs rather than to the content itself, and keep the content in a separate artifact store, because this makes receipts small and shareable while still allowing stored content to be checked against the record. (Addresses DR2, DR4, DR5.)

DP3 (Decide deterministically and record the decision). Implement provider and model selection as a pure function of a capability catalog, the request, and the policy, and record the selected route in the receipt, because a decision that can be recomputed can be audited. Treat output-policy violations as terminal, so that a run is never retried until a banned output passes. (Addresses DR3.)

DP4 (Redact before signing, verify before loading). Apply redaction before canonicalization and signing, so that matched sensitive values never enter the signed bytes in clear form, and verify a receipt before loading any referenced artifact, so that a forged receipt cannot trigger side effects. (Addresses DR1, DR2, DR4.) A hash in the signed bytes hides a value only when that value is hard to guess. A low-entropy value, such as a social security number or a yes/no output, can be confirmed from its unsalted hash by dictionary search. Salted or keyed commitments would close this gap and are left to future work.

5 Retrospective Review of Actual Refund Disputes

A refund dispute becomes an accountability problem when an organization must show what it told a customer, what its terms allowed, and what it later did about the money. Each question is answered by a different record, and the records are easy to conflate: an offer is not a credit, a temporary credit is not a refund, and an order to pay is not a payment. We use four published refund decisions to examine whether an AI-assisted review can keep these records apart and leave evidence that someone else can check.

5.1 Moffatt v. Air Canada

Moffatt v. Air Canada is the central case (Civil Resolution Tribunal, 2024). In November 2022, after their grandmother died, Jake Moffatt asked the chatbot on Air Canada's website about bereavement fares. The chatbot said that a ticket could be submitted for a reduced bereavement rate within 90 days of issue, even after travel. The Air Canada webpage it linked to said that the policy did not apply once travel was complete (Civil Resolution Tribunal, 2024, paras. 15–17). Relying on the chatbot, Moffatt booked two one-way flights and applied within the stated window, but later learned from Air Canada employees that retroactive applications were not permitted. An Air Canada representative later acknowledged that the chatbot had used "misleading words" (Civil Resolution Tribunal, 2024, paras. 2, 18–22). The Civil Resolution Tribunal found negligent misrepresentation. It rejected the airline's suggestion that the chatbot was responsible for its own statements, holding that the chatbot was part of the website and that the airline was responsible for all information on it (Civil Resolution Tribunal, 2024, paras. 27–32).

For this study, the decisive feature of the case is evidential. The customer's screenshot brought the chatbot's words into the proceeding (Civil Resolution Tribunal, 2024, para. 15). Air Canada supplied a screenshot of part of its webpage but no information about the nature of its chatbot (Civil Resolution Tribunal, 2024, paras. 14, 16). It relied on its tariff but did not provide the relevant portion, so its contractual argument failed (Civil Resolution Tribunal, 2024, para. 31). It also claimed a CAD 200 goodwill coupon as a set-off, but provided no evidence that Moffatt had accepted the coupon (Civil Resolution Tribunal, 2024, para. 41). The decision does not show whether the airline held better records; it shows that the airline did not produce them. An organization that cannot show what its automated channel said, and under which terms, is left with the customer's account of events. This is the kind of gap that run-level evidence is meant to narrow. Lattice played no part in this dispute, and nothing here shows that it would have changed the outcome.

The decision also marks the boundary between an order and a settlement. Paragraph 44 orders Air Canada to pay CAD 812.02 within 14 days: CAD 650.88 in damages, CAD 36.14 in pre-judgment interest and CAD 125 in tribunal fees. The total is therefore not a ticket refund alone. Paragraph 47 explains that a validated copy of the order can be enforced through the Provincial Court of British Columbia (Civil Resolution Tribunal, 2024, paras. 44–47). Neither paragraph is evidence of payment, and a decision issued on 14 February 2024 cannot record what happened after that date. Proof of settlement would be a different record, such as a payment confirmation or the claimant's acknowledgment, and the published decision contains none. The review task therefore asks whether the decision establishes payment, and the rubric treats "no settlement evidence" as the correct answer.

Published records are not necessarily self-consistent either. According to the tribunal, Moffatt paid CAD 1,630.36 for the two flights, and damages are calculated from that figure (Civil Resolution Tribunal, 2024, paras. 34, 40). However, the two fares it records sum to CAD 1,640.36 (Civil Resolution Tribunal, 2024, para. 18). The record cannot tell a reviewer which figure matches the actual charges; it can only show that they disagree. A review packet that ties each extracted value to the page it came from makes the disagreement visible, whereas a bare answer would hide it.

5.2 Three ombudsman decisions

Three decisions of the United Kingdom Financial Ombudsman Service present the same distinctions in other forms. In DRN-4377241, a booking website had registered its own email address with the airline, so the customer could not complete the airline's verification step for a refund, and the website declined to help. The ombudsman directed the customer's bank to refund GBP 77.44 with interest. It also directed the bank to pay GBP 100 compensation and refund GBP 19.41 in interest and fees, in each case only if it had not already done so (Financial Ombudsman Service, DRN-4377241). DRN-4148727 separates three sums that a hurried summary could merge: an approximately GBP 126 chargeback refund to the customer, GBP 958 that the airline later paid to the travel agency, and GBP 150 that the bank paid for service failings. The complaint was not upheld (Financial Ombudsman Service, DRN-4148727). In DRN-3785437, the bank temporarily credited GBP 13 while pursuing a chargeback and reversed the credit when it stopped (Financial Ombudsman Service, DRN-3785437). Each decision asks the consumer to accept or reject it by a stated date, and none records what happened afterwards.

5.3 The executed review

Figure 1 shows the executed workflow. The application fingerprints each PDF, extracts its pages and gives every page an identifier. It passes all pages of one decision, a fixed review task and an execution policy to Lattice. The task asks five factual questions per decision, including whether the decision establishes payment. Expected answers and accepted pages are not part of the prompt. A pinned free model returns structured answers, each with the identifiers of its supporting pages, together with a short review note. Lattice checks the response format and signs a terminal receipt that commits to the task, policy, serialized request and output. Afterwards, a separate verifier process checks the signature, recomputes the hashes and confirms that every page appears in the recorded request. A frozen rubric scores values and page references. These checks answer different questions: a valid receipt can preserve a wrong answer, and a correct answer is unusable if it cannot be parsed. Table 3 lists what each step retains and where its evidence stops.

The review is retrospective. The models read decisions whose outcomes are stated in the text, so the task is careful extraction rather than prediction. The four records were chosen for their evidential features rather than sampled, are public and may appear in model training data, and are published accounts rather than the companies' own systems or exhibit files. No refund was approved or paid in this experiment, and no disputes reviewer used the resulting packets.

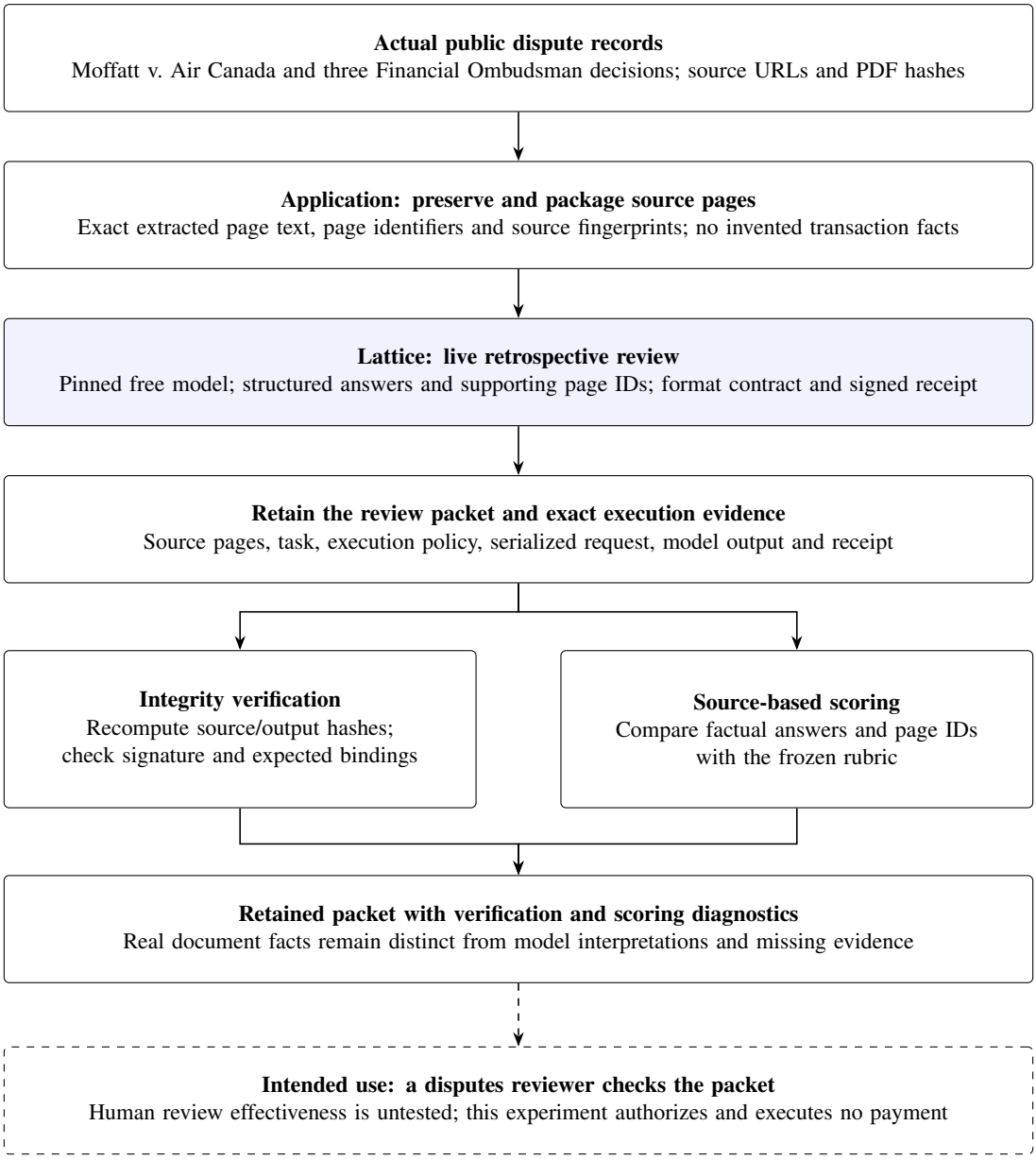


Figure 1: Executed retrospective review of actual published refund-dispute records: Moffatt (Civil Resolution Tribunal, 2024) and Financial Ombudsman Service decisions DRN-3785437, DRN-4148727 and DRN-4377241. Solid arrows show executed preparation, live inference and technical checks. The dashed final step is intended human use, not an observed practitioner study. Full decisions, including their outcomes, are supplied; this is document review, not prediction of a legal decision. Diagram prepared with OpenAI Codex; drafting model/version unconfirmed, pending author disclosure review.

Table 3: Actual public-record review: retained evidence and limits

Step / responsible role	Actual retained evidence	Implemented check	Remaining boundary
Acquisition / application operator	Four published PDFs, URLs, PDF hashes and exact extracted pages	Match PDF fingerprints, page counts and page-text hashes	Published decisions are not original company systems or complete exhibit files
Execution / Lattice operator	Task, page artifacts, execution policy, serialized request and model output	Confirm all supplied pages occur in the recorded provider request; bind task, policy and request	Request commitment does not prove provider consumption or honesty
Review packet / model and application	Structured factual answers, page references and review note	Format contract; separate frozen factual-value and supporting-page rubric	Source-page coverage is not full entailment; free-text review notes and legal correctness ungraded
Integrity / separate verifier process	Receipt, public key and locally retained input/output/request bytes	Verify signature, expected bindings and recomputed content hashes	No proof of truth or real-world completeness; no model re-execution
Organizational use / intended disputes reviewer	Packet and diagnostics; explicit unknowns, such as unconfirmed settlement	No human-performance or payment-execution check in this experiment	A direction to pay is not a completed payment; reviewer usefulness remains unevaluated

Worked source. In Moffatt, the published order totals CAD 812.02; the decision supplies no confirmation that this order was subsequently paid (Civil Resolution Tribunal, 2024, para. 44). The rubric distinguishes that direction from settlement. Source PDFs and exact request bodies remain locally retained; published bundles contain outputs, receipts and source/request fingerprints. Earlier invented-case tests are retained separately in Appendix B.

6 The Artifact: Lattice

Lattice is an open-source TypeScript library and command-line tool (Turlapati, 2026). A developer describes a run by its task, artifacts (text, structured data, images, audio, documents, or tool results), declared outputs with schemas, a policy, and an optional contract of output invariants. Figure 2 summarizes the run life cycle, and Table 4 maps principles to mechanisms.

Receipt contents. A signed receipt body holds the run identifier, issue time, signing key identifier, requested and observed model, selected route, token usage and cost, contract verdict and hash, input-artifact and output hashes, lineage root, and redaction manifest. It can also carry redacted tripwire evidence or route rejection reasons. Audience, nonce, expiry, parent, and closure fields remain optional. Schema v1.4 adds optional `taskHash`, `policyHash`, and `providerRequestHashes`. Current signed `ai.run` and public `ai.runAgent` results populate the task and effective-policy hashes using SHA-256 over canonical JSON. The ordered request list is included only when every executed adapter supplies the submission callback and at least one body was prepared; it includes prepared attempts that later fail. Each entry hashes the exact serialized UTF-8 body handed to configured fetch, after packaging. Transport headers and the endpoint URL are excluded from that hash, and raw content is not placed in the receipt. A custom fetch can subsequently modify those bytes, and the digest proves neither remote receipt nor model consumption. Legacy receipts and adapters without complete callback coverage remain visibly unbound. Verifiers accept optional fields when absent unless the caller supplies corresponding expected digests; a version number alone does not require their presence. Routing candidates, scores, and catalog version remain outside the signed record, so DR2 and

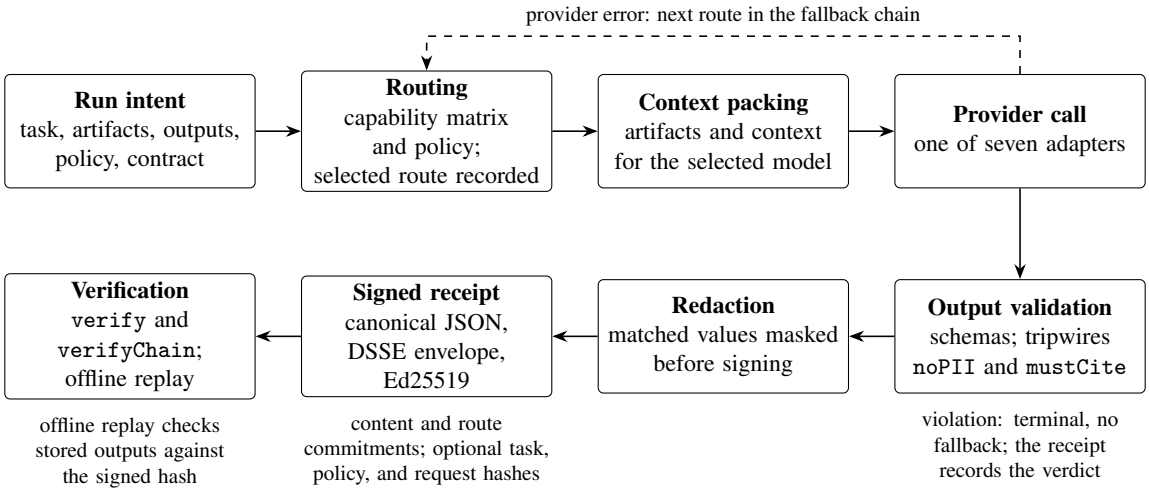


Figure 2: Run life cycle of an `ai.run` call in Lattice. Solid arrows follow one run. The dashed arrow is fallback on a provider error; an output-invariant violation ends the run. Current optional task and policy hashes commit to canonical inputs, while request hashes commit to serialized bodies handed to configured fetch

DR3 require separately retained evidence beyond these commitments.

Provider coverage. Seven adapters (OpenAI, generic OpenAI-compatible endpoints, Anthropic, Google Gemini, xAI, OpenRouter, and LM Studio) share a provider contract. Current ordinary and streaming builders deliver supported image values as native image parts, preserve JSON values, and reject unsupported or unsatisfied transports before submission. Binary image views retain their byte ranges; URL delivery remains subject to `noPublicUrl`. The earlier vision probe in Section 7.8 records the old builder, which dropped binary and object values or sent image data as text; it is historical evidence of the defect, not a measurement of the correction. Optional `generation.temperature` and `generation.maxOutputTokens` now flow through calls, agents, and crews into provider-specific wire fields. Native OpenAI uses `max_completion_tokens`, compatible gateways use `max_tokens`, and native Anthropic and Gemini use their own fields. Omitted settings preserve earlier defaults. The request digest records these serialized choices, but does not validate provider capability claims or prove that a model used an input. The router's capability catalog remains a point-in-time registry supplemented by maintained profiles.

Agent runs. Public `ai.runAgent` now returns a signed terminal receipt on successful and failed result variants when an intent or runtime signer is configured, independently of tracing. It commits to the original task, effective policy, terminal verdict, successful output when available, and the complete observed provider-submission sequence when supported. Its default signer-failure behavior returns `receiptError`; the configured throw mode raises an error. Per-iteration checkpoints remain a separate tracer facility. Crew execution retains its existing receipt accounting: a crew-start root, a completion receipt for the root and each successful child run, and a crew-close receipt. Thus N agents that each run once with successful children yield $N+2$ crew receipts; a failed child has no crew completion receipt. The crew-start hash covers

Table 4: Design principles and their instantiation in Lattice

Principle	Mechanism in Lattice
DP1 Receipt by default	A configured signer produces a receipt on terminal results of `ai.run` and public `ai.runAgent`. Agent intents can override the runtime signer; terminal receipts need no tracer. Per-iteration checkpoints remain separate. Crews return a crew-start receipt, completion receipts for the root and successful child runs, and a crew-close receipt. Receipts use RFC 8785 canonicalization (Rundgren et al., 2020), DSSE (Secure Systems Lab, 2021), and Ed25519 (Josefsson and Liusvaara, 2017; Bernstein et al., 2012). Key sets track active, retired, and revoked keys; version floors reject unsupported older schemas.
DP2 Commit to content	Receipts contain SHA-256 hashes of input artifacts and outputs and a Merkle root over artifact lineage (Merkle, 1988). Optional v1.4 fields add task and policy hashes and an ordered list of provider-request hashes. Content is kept separately. Monetary cost is serialized as an I-JSON string (Bray, 2015).
DP3 Deterministic decision	The router applies hard filters (modality, context window, structured output) and policy filters (allow and deny lists, privacy class, logging, latency, budget), then ranks survivors by a score combining cost, latency, and context headroom. Ties break on provider and model identifiers, not on catalog position. Output invariants (`mustCite`, `noPII`, `fieldFromTable`, `matches`) are evaluated by a pure function; violations are terminal and excluded from fallback.
DP4 Redact, then sign; verify, then load	Canonicalization is called only on the output of redaction. Materialization for offline replay verifies the receipt before any artifact is loaded. Input and output hashes are unsalted SHA-256.

the nested identifiers, not the whole specification, and completion receipts hash outputs. The single-agent extension does not retroactively add full task, tool, or policy commitments to these crew records. Current `verifyChain` enforces the closing member list; historical 1.6.x verifiers do not. `requireClosure` rejects missing closure, while `partial` permits only listed subsets with a closer. Crew work time ends before root completion and closure signing, but includes start and child signing. Failed mints are reported under the signer-error policy; a verifiable closing list names only receipts actually minted, so callers must also inspect `receiptError`.

External executors. An organization that already uses another orchestration framework can call an audit helper that mints the same receipt family from externally produced request, response, and output data. This supports incremental adoption (DR5), at the price of trusting the external executor's account of the provider call.

Command-line tool. The tool verifies receipts and receipt chains, prints structured advisories as warnings or information lines, checks stored artifacts and outputs offline against the signed hashes, compares receipts structurally, and gates regressions in cost and output consistency across a corpus of receipts in continuous integration. The library can also export run events through OpenTelemetry (OpenTelemetry Authors, 2026), and tool results from the Model Context Protocol (Model Context Protocol, 2025) can be recorded as artifacts.

Table 5: Historical evaluation of design requirements: experiments, criteria (Prat et al., 2015), and outcomes at the recorded revisions

Requirement	Experiment	Criterion	Outcome
DR1 Integrity	Tamper suite on 70 receipts; chain splicing; verifier opt-ins	Robustness	Single-receipt attacks 3570/3570; freshness binding and open-chain length are opt-in; closed crew chains fail on a missing member
DR2 Reproducibility	Record, persist, reload, and re-check 350 runs across 7 adapters	Consistency	Hash match 350/350; historical receipts bind artifacts and outputs; task binding was not tested; no live-behavior claim
DR3 Decisions and policy	1000 routing requests with an independent oracle; labeled invariants	Goal attainment	No policy violations (0); permutation-stable (1.0000); selected route signed, rationale not; PII and must-cite below
DR4 Minimization	Secret-leakage scan of signed receipts	Goal attainment	Marker secrets in signed bytes: 0 of 14 rows; low-entropy hashes can be guessed
DR5 Low cost	Latency over 1000 trials; receipt sizes	Efficiency	Signing delta 0.214 ms (small text) to 1.354 ms (1 MB image), p50

7 Evaluation

7.1 Evaluation design

The original artificial, summative evaluation (Venable et al., 2016) tested the instantiation against DR1–DR5. Its measurements and the earlier local/free-hosted recordings below remain historical results at their recorded revisions. They are not measurements of the later v1.4 binding, terminal-agent, transport, or generation-setting changes. A separately generated validated-evaluation fragment follows the historical recordings when available; no improvement is inferred merely from implementation changes. The harness is published with the artifact (`eval/run-all.sh`). In that artificial evaluation, provider calls were served by in-process fakes that return provider-shaped responses, so no network, credentials, or paid services were involved. Table 5 maps requirements to experiments, and Figures 3, 4, and 5 plot the personal-data check, the per-run latency, and the cost of receipt chains. Section 7.7 is separate: it records a local open-weight model and the other record systems that could be executed on the same machine (Table 6; Figures 6, 7, and 8). Section 7.8 adds free hosted models (Tables 7 and 8), public personal-data datasets (Tables 9 and 10; Figures 9 and 10), a citation dataset, and local gateways. Its other figures give Lattice processing share, signed latency, answer drift, and output binding on the live models (Figures 11, 12, 13, and 14), mutation results and observed capabilities across the record systems (Figures 15 and 16), and the gateway runs (Table 11; Figure 17). Synthetic data were generated with fixed random seeds. Hardware, the Node version, and the git commit are in Appendix A and in the environment files under `eval/results`, not in this narrative, because they are properties of the measurement machine. Tables in the appendix are generated from the JSON the harness writes: latency, receipt size, and chain cost (Tables 22, 23, and 24), and the replay, policy, tamper, and verifier checks (Tables 25, 26, 27, and 28). The prose below quotes those generated values through macros, so a number in the text is the number in the file.

7.2 Integrity under attack (DR1)

We generated 70 receipts from runs across all seven adapters and applied the mutations in Table 27. Of the cases the harness classifies as attacks, 3570 of 3570 were rejected. Body edits failed signature verification whether the attacker kept the original signature, re-canonicalized correctly, or re-signed with an untrusted key under the legitimate key identifier. Non-canonical encodings were rejected as canonicalization mismatches. A valid receipt presented a second time is accepted: the verifier keeps no replay cache. Audience, nonce, and a caller-supplied version floor are checked only when requested. `expiresAt`, when the issuer set it, is enforced on every verification. Presenting the same receipt again is therefore not, by itself, a failed verification. What the verifier did not check is reported as structured advisories, each with a code and a severity. A warning means the receipt is bound to an audience or a nonce that the caller did not check. An information advisory means the receipt carries no audience, nonce, or expiry, or sits below the current schema version. The command-line verifier always prints warnings and prints information lines unless `-quiet` is set. Figure 15 applies comparable mutation classes to the other record systems of Section 7.7.

Chain checks are separate from the single-receipt `VerifyErrorKind` taxonomy. `verifyChain` reports its own error kinds (`VerifyChainErrorKind`): an invalid link, no root, multiple roots, an orphan link, a run-identifier mismatch, a duplicate content identifier, a count mismatch, or a root mismatch. Reordering the receipt array is not an error, because the verifier follows parent links rather than array position. Dropping a middle link, splicing a foreign link, or replacing the root fails that check. Truncating the tail of an open chain does not. The result is success plus a tail-truncation warning unless the caller passes `expectedCount`. A chain whose root declares `chainClosure` fails with a count mismatch when a leaf or the closing receipt is missing, with no expected count, and a malformed closing receipt fails as an invalid link. `requireClosure` turns a missing closure declaration into a failure, and `partial` accepts a disclosed subset only when the closing receipt is present and lists every presented receipt, and warns when some listed members were not presented. Table 28 records these checks together with the advisories, crew receipts, and signer-failure modes. 71 of 71 checks passed. Two of them run the released 1.6.0 verifier on a closed crew chain and confirm that it accepts the chain without enforcing closure.

7.3 Reproducibility (DR2)

For each of the seven adapters we recorded runs against provider-shaped fixtures, wrote the receipt and content-addressed artifacts to disk, reloaded them, verified the receipt, materialized a replay envelope, replayed offline, and recomputed the output hash. 350 of 350 runs verified, and 350 reproduced the signed output hash. Re-executing each run against the same fixture matched in 350 cases, and a one-character change to a recorded output was detected in 350 cases (Table 25). These results show that the record is internally consistent. Offline replay does not call a model: `replayOffline` returns the outputs the caller supplies, and the check is that they hash to the signed output hash. The result is therefore a signed output binding, which other signed record systems also provide (Table 6). On the hosted rerun, every replayed receipt verified and a one-character edit was detected, and the signed output hash matched the kept answer except on trials that stored an empty answer after a provider failure (Figure 14). That mismatch is not a signature or chain failure. The check does not show that a live model would reproduce its output, and repeated

live calls often returned different answers (Figure 13). Those historical bindings covered input artifacts and outputs only, with no task commitment. Current optional v1.4 fields extend that boundary, but do not turn the historical replay counts into tests of task or policy binding. In a crew chain, the crew-start receipt hashes the nested agent tree (`{kind: "crew-start", agents: {id, children}}`), each completion receipt hashes its agent's output, and the crew-close receipt hashes its member list.

7.4 Inspectable decisions and policy enforcement (DR3)

Routing. Using a synthetic catalog of 40 model capabilities and 1000 random requests with random policies, repeated decisions were identical at rate 1.0000. An independently implemented oracle found 0 selected routes that violated provider allow or deny lists, privacy class, logging, latency, or image-modality constraints, and 0 cases in which the router reported no route although a feasible candidate existed. Reordering the catalog changed the selected model at a same-selection rate of 1.0000. Catalog position is not part of the score, and ties break on provider and model identifiers. The receipt signs the selected route and the requested and observed model. In the historical experiment it did not sign the policy. Current task and policy hashes permit expected-content checks, while the policy itself, candidate list, scores, and catalog version still require separately retained records to recompute the decision.

Personal-data invariant. The following rules and scores describe the detector revisions used by the historical experiments. They cover four types, and each rule trades recall for precision in a stated way. Email addresses match as written. A dashed United States social security number (SSN) matches as written. Spaced and undashed nine-digit forms match only when an SSN cue such as `ssn`, `ss#`, or `social security` is the nearest word. Separated United States phone numbers match as written. A plain ten-digit run matches only when it is a valid North American number and the nearest word is a phone cue (`phone`, `tel`, `call`, `mobile`, `cell`, or `fax`) or a `+1` or `1-` trunk prefix precedes it. A payment card number is any Luhn-valid run of 13 to 19 digits, except runs that start with 0 or 1 and 13-digit runs that start with 978 or 979, which are ISBN-13 numbers. An SSN, phone, or card match is suppressed when an identifier label (`order`, `ord`, `invoice`, `inv`, `sku`, `ref`, `po`, `isbn`, `pn`, or `txn`) is the nearest token on the same line, joined to the number only by whitespace or by `:`, `#`, `=`, `no.`, `number`, or `is`, unless the number is a phone number with a trunk prefix. Labels do not suppress email addresses. Words such as customer, account, and ticket are not labels. The trade-offs follow from these rules. A real SSN written as `order 123-45-6789` is not flagged. A bare nine- or ten-digit number without a type cue is not flagged, whether or not it is personal data. About one in ten random digit strings passes the Luhn check, so a text that lists k numeric identifiers of 13 to 19 digits is flagged as holding a card with probability near $1 - 0.9^k$. The detectors see only strings, so a value stored under a key such as `ssn` is judged by its digits alone.

We built a labeled set of synthetic strings, 100 per label. In-scope positives cover each form the detectors claim, support-ticket text in which a number follows words such as customer, account holder, or patient ID, CSV and markdown rows, and card numbers from several networks. Out-of-scope personal data are names with addresses, international bank account numbers, United Kingdom phone numbers, IP addresses, dates of birth, and spelled-out email addresses. Negatives include order identifiers and stock-keeping units

shaped like SSNs or phone numbers, 16-digit references, ISBN-13 numbers, and uncued numbers such as timestamps, byte counts, and bank routing numbers. On in-scope types plus all negatives, `noPII` reached precision 1 and recall 0.9429 ($n = 5500$, false positives 0, false negatives 200). The false negatives come from forms the rules leave out by design, such as a plain ten-digit number after "call the customer at" and an undashed SSN in a CSV row with no SSN cue. The 1000 uncued numeric negatives produced 0 false positives. Counting every personal-data label, including types the detectors do not claim, precision was 1 and recall 0.8083 (Table 26, Figure 3). The cases were chosen with the rules in view, and the identifier-shaped negatives carry the labels the rules key on, so these figures show that the rules behave as specified. They are not prevalence-weighted estimates and not estimates of precision on real text. On public datasets, SSN and phone precision and recall were far lower, because passport numbers and other identifiers share the same digit shapes (Section 7.7, Figure 7, Tables 9 and 10, Figures 9 and 10). Those public-data rows were measured with the rules at commit `a8cc091`, which differ from the rules above as Section 7.7 describes.

Citation invariant. The `mustCite` invariant checks that an output cites a named artifact. By default it reads only structured lists: a top-level `citations` or `evidence` list whose string entries, or whose objects' `source` field, name the artifact. Matching is case-sensitive unless case folding is requested, and other object keys can be named. If the top-level list is missing or omits the artifact, nested `citations` and `evidence` lists are searched, unless the contract declares a path. The default never reads prose. Three options extend it, and each is added to the contract only when set, so the hash of an existing contract does not change. With `inline`, a bracketed or parenthesized group in the prose whose content, or one comma- or semicolon-separated part of it, equals the artifact name also counts; it does not override a non-empty structured list that omits the artifact. With `rejectUncited`, a matching list entry is rejected when the prose contains citation-like square brackets, such as `[1]`, `[1-3]`, or `[Smith 1999]`, and none of them resolves to the artifact. Links, footnote markers, and code are not treated as citations. `textPath` limits these prose checks to one field. The invariant agreed with the labels in 36 of 36 constructed cases: 22 of 22 for the default and 14 of 14 for the options. The cases include numeric markers, links, and path-scoped outputs, and they were also chosen with the implementation in view. ASQA long answers are prose without citation lists, so the default accepts a gold title in none of them. The ASQA count in Section 7.7 was measured at commit `a8cc091`, where an inline marker containing the name counted by default. The check is syntactic. It does not judge whether the cited artifact supports the claim (Lewis et al., 2020).

7.5 Data minimization (DR4)

We placed marker secrets in the task, an input artifact, the provider credential, and the model output, then scanned the decoded signed payload, with and without a `noPII` invariant. Marker hits in signed bytes were 0 of 14 rows (Table 26 does not repeat the per-adapter log; the raw rows are in `eval/results/policy.json`). Receipts carry hashes of the input artifacts and the output. In that historical recording, neither the task text nor the credential appeared in the signed bytes, in clear or as a hash. Current receipts add an unsalted task digest and may hash request bodies containing sensitive content; credentials in transport headers are outside that digest. When the invariant fires, the in-process tripwire result still includes the matched substring so the

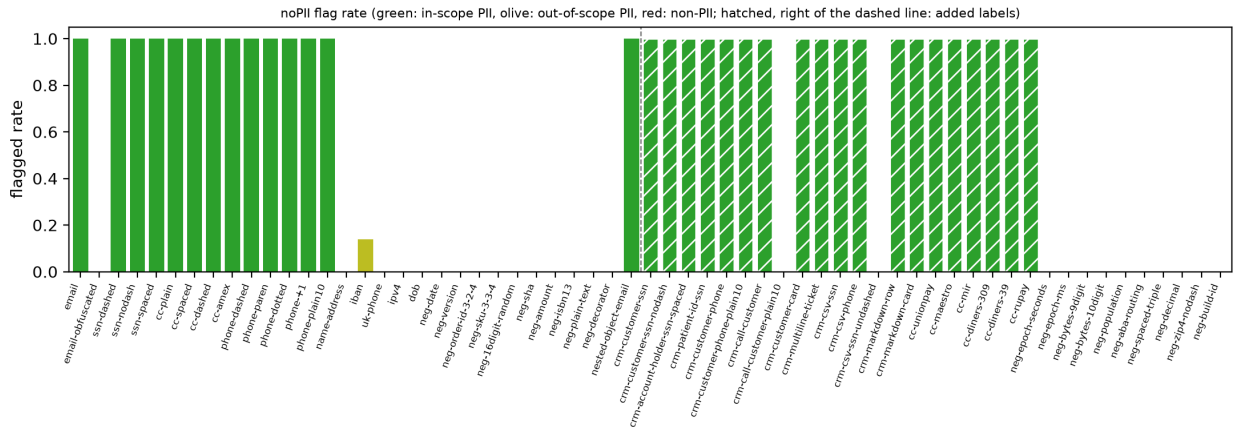


Figure 3: Share of samples flagged by the `noPII` invariant, by label. Green: in-scope personal data. Olive: out-of-scope personal data. Red: non-personal data. Plotted from `eval/results/policy.json`

caller can act on it. The value that is signed is a detector name, a length, and a mask, so the signed bytes hold the masked form that the redaction manifest describes. The scan checks for the marker strings themselves. It does not test whether a hash can be inverted: a low-entropy input or output, such as an SSN or a yes/no answer, can be confirmed from its unsalted hash by dictionary search (Section 4, DP4). Among the record systems compared in Section 7.7, Figure 8 shows which kept the plaintext transcript and which kept only digests or ciphertext when wrapping the same recorded request and response bytes.

7.6 Adoption cost (DR5)

Table 22 and Figure 4 report latency over 1000 trials after warm-up, against a mocked provider. Signing added 0.214 ms at the median for a 1 KB text input and 1.354 ms for a 1 MB image (signed run minus unsigned run). Verification of the small-text receipt was 0.172 ms at the median. The receipt grows with the number of input hashes rather than with payload size (Table 23). Linked receipts and chain verification scale with depth (Table 24, Figure 5). Because the provider was a zero-latency fake, these figures are the runtime's own cost. Sections 7.7 and 7.8 measure that processing time again on a local model and on hosted models, with provider latency and rate-limit waits kept on separate clocks (Figure 6, Tables 7 and 8, Figures 11 and 12).

7.7 Local open-weight runs and other record systems

This subsection preserves historical measurements at the revisions recorded below. Current transport, generation controls, single-agent receipts and scoring differ; these figures are not a controlled before/after comparison. The artificial evaluation mocks the provider; the local runs here used a live model. Ollama 0.40.7 served `llama3.2:3b` (Llama 3.2, 3.2B, Q4_K_M, digest `a80c4f17acd5`, 2.0 GB) on the same host as the other measurements (Node v24.22.1, 4 vCPU, 15.63 GB RAM, no GPU). Lattice called it through the OpenAI-compatible adapter with no API key and an explicit base URL ending in `/v1`. `createOpenAIProvider`, whose default base URL builds a chat URL without `/v1`, was not exercised. At that historical revision the

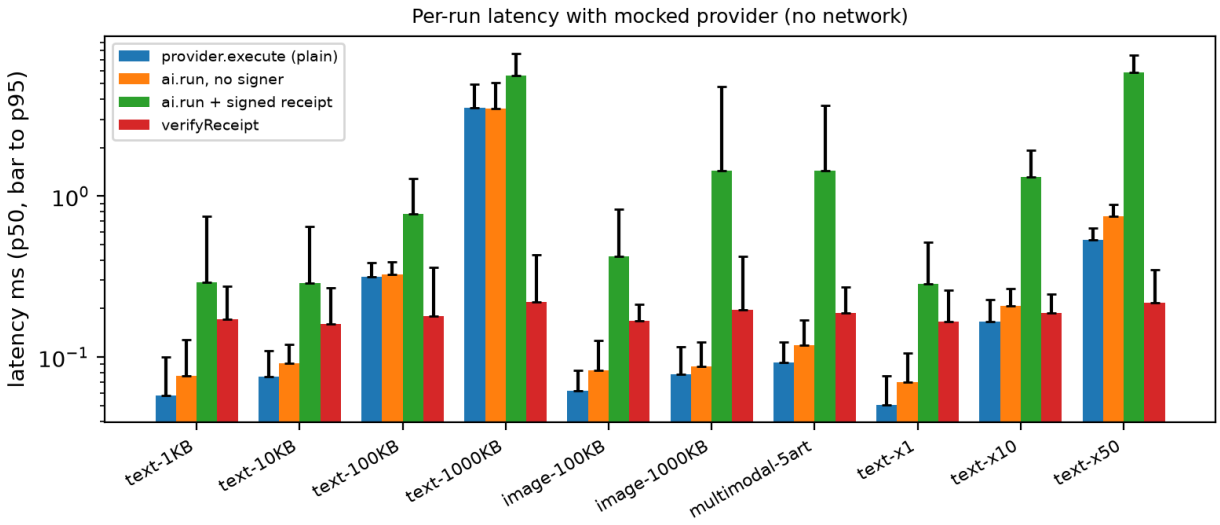


Figure 4: Per-run latency with a mocked provider across input configurations (log scale; bars show the median, whiskers extend to the 95th percentile)

adapter did not send temperature or a token cap, so a fetch wrapper added both to every request. The Lattice numbers are therefore for a request the adapter does not build on its own. No vision model was loaded. No hosted provider key was present.

On the one-line prompt "Reply with the single word pong," temperature 0, five paired trials, the median plain HTTP call was 278 ms. A signed Lattice run of the same task was 559 ms and an unsigned Lattice run was 745 ms (Figure 6). The first plain call, 4845 ms, is a cold start and pulls the mean up; the median does not. The signed median is lower than the unsigned median. That gap is generation noise, not a claim that signing is faster. What is stable is the time spent outside the provider HTTP call: 0.439% of end-to-end latency at the signed median (minimum 0.335%, maximum 2.469%). The Lattice prompt body was 382 characters against a one-line plain prompt, so most of the extra wall time is a larger prefill, not the signature. The short signed receipt was 996 bytes.

A refund-policy question with a `mustCite` invariant passed on all three temperature-0 runs and all three temperature-0.7 runs. These verdicts come from the `mustCite` rules at commit `a8cc091`, where an inline `[name]` or `(name)` marker containing the artifact name counted as a citation by default. The current default reads only a structured citation list, so these answers, which cite the clause only inline, would fail it unless `inline: true` is passed. Each temperature produced one identical answer across its three trials (n=3). The two temperatures did not produce the same sentence. All 6 signed receipts verified. For each, the signed output hash matched the kept answer and changed after a one-character edit (6/6). This checks the output binding, not a re-run: `replayOffline` returns the outputs the caller supplies. A synthetic support ticket containing an email, a US phone number, and a dashed SSN was flagged by those three detectors. The model's answer contained none of them, the `noPII` invariant passed, and the signed receipt did not contain the planted strings. A two-step agent called `lookup_policy` once and answered; the returned agent object carried no receipt. A two-agent crew completed in 3 iterations with 4 receipts. These local-model rows do

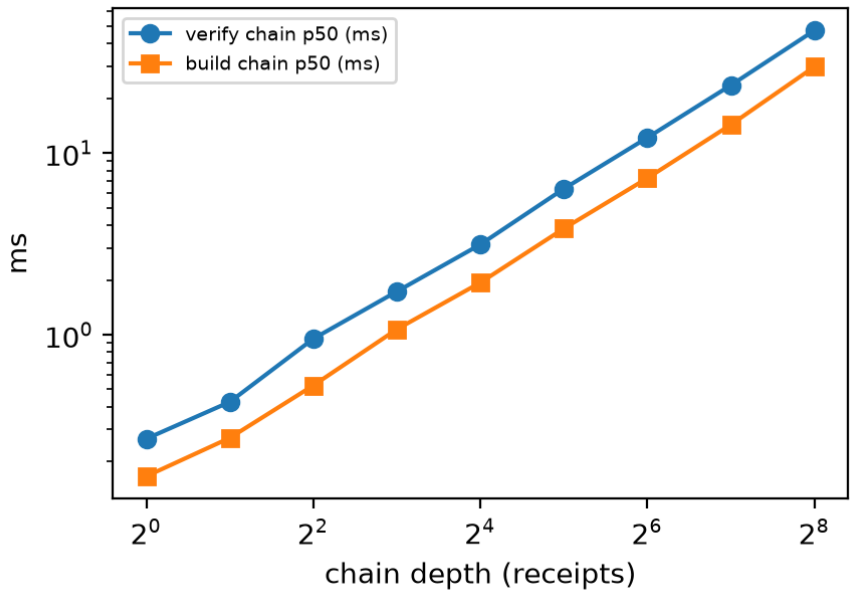


Figure 5: Build and verification time of linked receipt chains by depth

not name a library commit. They were recorded before the crew-receipt and chain-closure changes and were not rerun on this revision.

The public sets are harder than the synthetic harness, and the gap is the result (Figure 7). ai4privacy/pii-masking-300k (Ai4Privacy, 2024) English validation (revision `c8c77895a005`, 7,946 documents; academic, non-commercial, no redistribution) gives email precision 0.955, recall 0.979, and F1 0.967 (1533 documents with a gold email). US SSN precision is 0.988, recall 0.425, and F1 0.594. US phone precision is 0.453, recall 0.369, and F1 0.407. Of the 8 SSN false positives, none overlap a passport span. The cue rule keeps a 9-digit passport from matching when no SSN cue is the nearest token. Of the phone false positives, 532 overlap a social-security span. There is no gold credit-card span in this file; the card detector fired on 40 documents anyway. Among 1,070 US-shaped SSN spans, the historical first-hit API overlaps 646. It misses 348 because the first match is elsewhere and 76 because it matched nothing. This is not all-span extraction recall. Gretel English finance test (Gretel.ai, 2024) (Apache-2.0, revision `7b844d167385`, 2,891 rows) is the same pattern at smaller n: email precision 0.522, recall 0.934, F1 0.670; US SSN precision 0.984, recall 0.857, F1 0.916; US phone precision 0.868, recall 0.687, F1 0.767. ASQA dev (Stelmakh et al., 2022) (Apache-2.0, revision `084060f16b46`, 948 answers) is a loss for `mustCite`. A gold Wikipedia title occurs as a substring in 525 answers. Only 2 answers use a square-bracket citation. With `inline: true`, the invariant accepts at least one gold title in 4 of 948. The structured default, which never reads the answer text, accepts a gold title in 0. A bracket or parenthesis counts only when its content, or one comma- or semicolon-separated part of it, equals the title. The check is syntactic. It does not score whether a citation supports the claim. These dataset rows were measured with the `noPII` detectors at commit `ae99fb5`.

Table 6 wraps the same recorded request and response bytes. Lattice's receipt and Sello's encrypted receipt (mock transparency log, not the public log) omit the planted strings. An AFR log with disclosure

Table 6: Same recorded Ollama runs, wrapped by each system that ran. Wrap time excludes model generation. Wrap ms is one cold run per system, not a warmed median, and the scopes differ: Lattice and Sello are in-process JavaScript calls summed over three records, cosign and the NovaFabric capsule are one CLI invocation each, including process start and key loading, Janus reports whole milliseconds from its own timer, and the JSON log time is serialization only. PII in artifact reflects what the harness gave each system; where it is yes, the byte count includes the plaintext transcript. Signed output binding is yes when the record is signed and the system's own verifier rejected a byte edit. TrustTrack and Langfuse were not executed. The cosign run uploaded to the public Rekor log and verified online.

System	Status	Wrap ms	Bytes	Offline verify	Byte edit detected	Key assumption	PII in artifact	Signed output binding
Lattice receipt	measured	5.923	3210	yes	yes	other key rejected	no	yes
JSON log	baseline	0.033	1045	no	no	—	yes	no
Hash chain	baseline	0.082	1531	yes	yes	rebuild accepted	yes	no
OTel GenAI spans	measured	0.228	1366	no	no	—	yes	no
Agent Flight Recorder	measured	0.797	2328	yes	yes	mock anchor only	yes	no
AFR disclosure	measured	1.907	3357	yes	yes	mock anchor only	no	no
NovaFabric HMAC	measured	0.077	1432	yes	yes	shared secret re-signs	yes	no
NovaFabric capsule	measured	1373.053	21288	yes	yes	self-asserted key	yes	yes
cosign sign-blob	measured	4916.329	1319	online	yes	local key, Rekor upload	yes	yes
in-toto link	measured	34.149	888	yes	yes	functionary key	no	yes
Sello receipt	measured	9.336	1359	yes	yes	mock log	no	yes
Janus evidence log	measured	5	1723	yes	yes	writer key	yes	yes
TrustTrack	paper only	—	—	—	—	—	—	—
Langfuse	not run	—	—	—	—	—	—	—

encryption also omits them; the default AFR log contains them. Janus's local evidence appender stores the payload in the clear and the byte edit fails `verify.SegmentDir`. A sealed NovaFabric capsule (self-asserted ECDSA key, no RFC 3161 timestamp) fails `nova verify` after a newline is appended to stdout, and the capsule contains the planted text. `nova validate` on that edited capsule still exits 0. cosign `sign-blob` (The Sigstore Authors, 2021) with a local key and no Fulcio certificate detects a byte edit and signs the plaintext log. That run kept cosign's default transparency-log upload, so it wrote an entry to the public Rekor log. Its 4,916 ms wrap time includes that round trip, and `verify-blob` checked the log online. in-toto's DSSE link (Torres-Arias et al., 2019) detects a byte edit and does not embed the plaintext. A hash chain detects an edit that leaves the stored hashes untouched and accepts a rebuilt chain, because nothing outside the file anchors the head. NovaFabric's HMAC record has the same shape with a shared secret: an edit fails, and a re-sign with that secret passes. OpenTelemetry GenAI spans and a plain JSON log have no integrity check; both store the transcript. TrustTrack has no implementation we could run. Langfuse was not started: this host has no Docker daemon. Lattice, the NovaFabric capsule, cosign, in-toto, Sello, and Janus sign the record and reject a byte edit; Table 6 lists that as signed output binding.

The wrap times are single cold samples with different scopes, so they order the systems only roughly. The Lattice total of 5.923 ms is three `createReceipt` calls, and the first call is 5.584 ms of it; Sello's first call is 7.259 of its 9.336 ms. The byte column mixes scopes too (Figure 8): the JSON log, the hash chain, OTel spans, AFR, the NovaFabric HMAC record, the NovaFabric capsule, cosign, and Janus keep the plaintext transcript, while Lattice, AFR disclosure, in-toto, and Sello keep only digests or ciphertext.

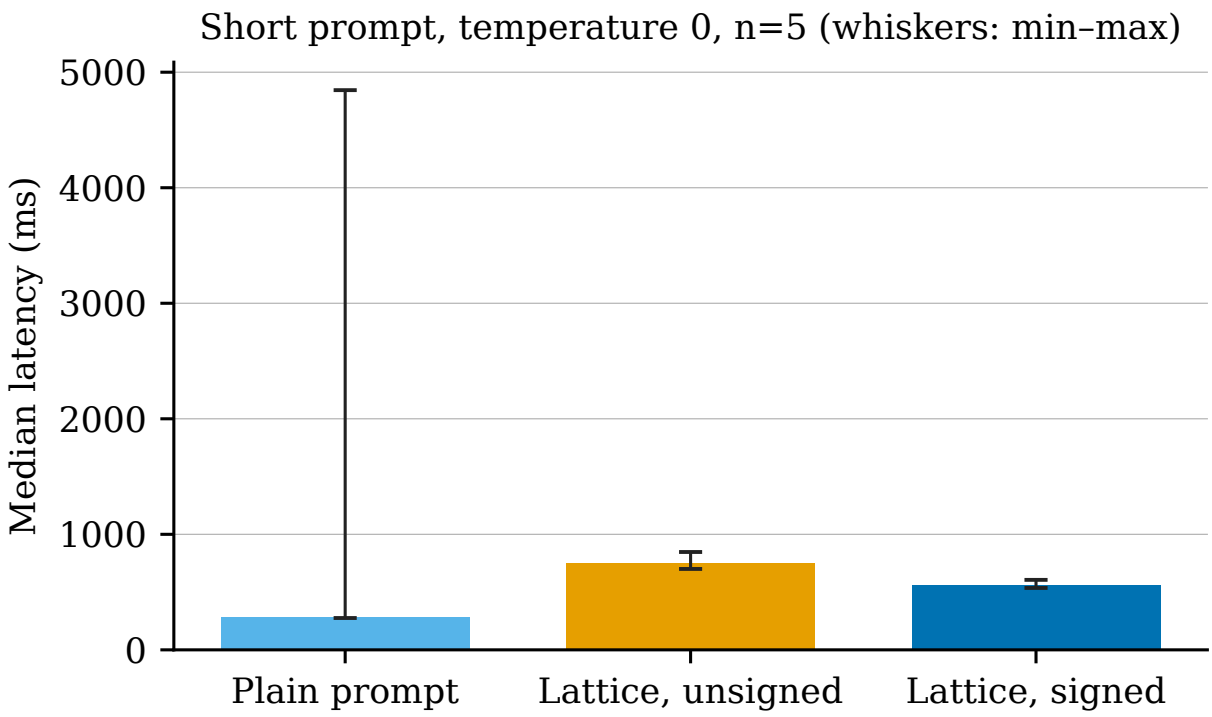


Figure 6: Median latency of a one-line completion. Whiskers run from the minimum to the maximum of five trials. The plain arm sends only that line. The Lattice arms send the adapter's packed prompt.

7.8 Broader comparison

These are historical recordings, with library revisions and scorer differences stated below. Current implementation results are reported separately.

This subsection adds the detector, citation, and gateway measurements from the local harness, and a hosted-model rerun recorded on the author's machine.

Lattice's OpenAI-compatible adapter called `https://openrouter.ai/api/v1`. The catalog check found 7 of the requested `:free` ids, and the listed prompt and completion prices were 0. Completed model rows: 7. Upstream HTTP 429 responses were retried. Pacing sleeps, that backoff, and the HTTP time of a failed attempt are excluded from Lattice time and from end-to-end time, so end-to-end time is Lattice processing plus the provider time of the attempt that returned. Provider time includes waiting for the response headers and reading the body. Completion tokens are capped at 8192. At that revision the adapter sent no temperature, `max_tokens`, or reasoning field. A fetch wrapper adds temperature and the token cap to every request, and adds `reasoning.effort=low` for models whose catalog lists a reasoning parameter. A 400 that rejects that field is retried once without it. The Lattice rows therefore describe requests the adapter does not build on its own. Table 7 uses the provider string returned on the chat response. A drift cell is distinct successful answers, successful calls, and attempts. A RAG call succeeds when the run returns an answer that passes `mustCite`, which was called with `inline: true` because the answers cite the clause inline. Output binding counts replayed receipts whose signed output hash matched the kept answer, over every replay row,

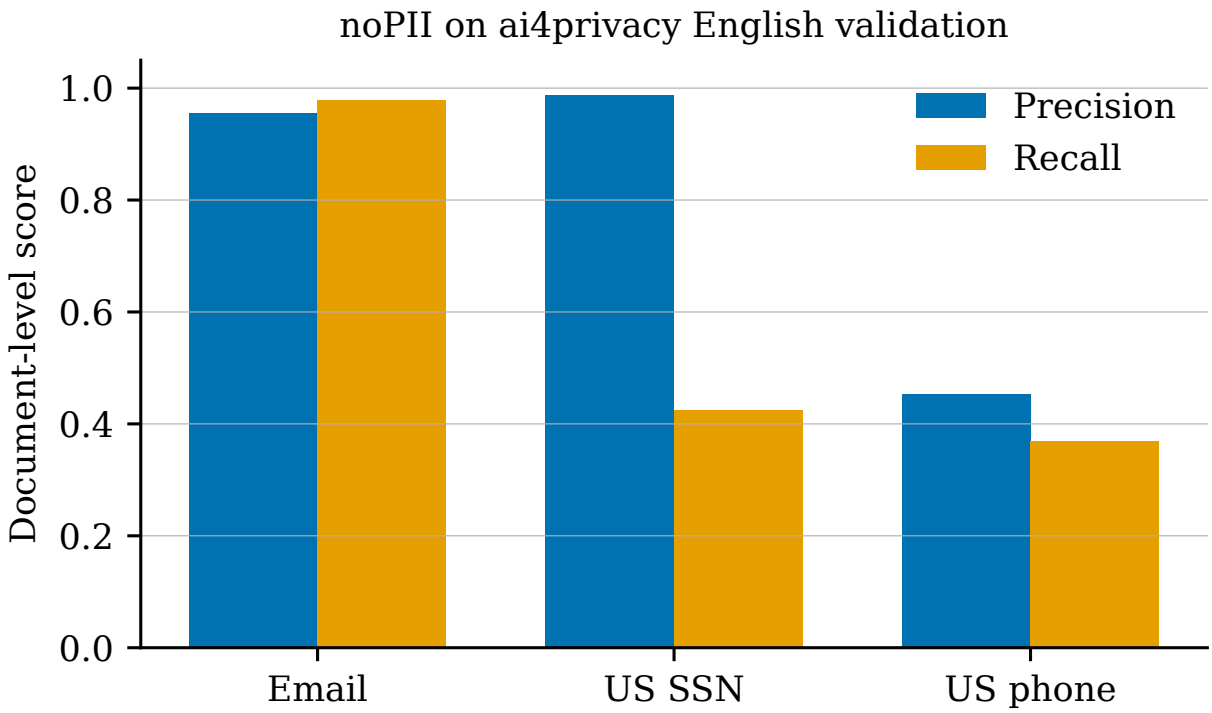


Figure 7: Document-level precision and recall of the default `noPII` detectors at commit `ae99fb5` on ai4privacy English validation. US phone false positives still overlap social-security spans. US SSN false positives no longer overlap passport spans; recall is the remaining loss.

including calls that kept an empty answer. The local row is the Ollama run in Section 7.7 (drift n=3, overhead n=5). Hosted drift used n=10 when the call returned. This hosted recording is library commit `c5f79b0`, started at 2026-10-26 14:13 CDT (2026-10-26T19:13:36Z), on the author's machine. The local Ollama rows, the public-dataset rows, and the gateway rows are earlier recordings and are not from that machine. The plan called 5 short-prompt trials per model and replayed the RAG receipts (20 per model that returned them). Only free OpenRouter models were called. The single-agent stage and the image stage were left out of this recording. On the local Ollama run, the median time outside the HTTP call was 2.4 ms (p95 15.0 ms, 0.439% of that run's signed wall time). That localhost call returns the body with the headers, so the gap is processing time. Across hosted models with a successful signed short prompt, the median Lattice processing time was 1.66–3.03 ms, or 0.066–0.265% of end-to-end time (1.7 to 3.0 ms and 0.07 to 0.27% at the reported rounding). Table 8 gives the median and p95 in milliseconds and as a percentage. With five trials, p95 is the largest trial. The percentages are recomputed from the per-trial phase times as Lattice milliseconds over Lattice milliseconds plus the provider time of the attempt that returned. The `signedLatticePct` field stored in the previous `openrouter.json` also counts the HTTP time of retried attempts, so the comparison uses the recomputed rows for both runs. Lattice time includes `createAI()` construction and the fetch wrapper's JSON handling, so it is an upper bound. On the previous recording those medians were 1.10–1.69 ms, or 0.032–0.218% of end-to-end time. The new medians are higher. Liquid's percentage p95 is 0.72%, the highest hosted p95 in this run. The packed Lattice prompt added 62–81 prompt tokens at the median versus

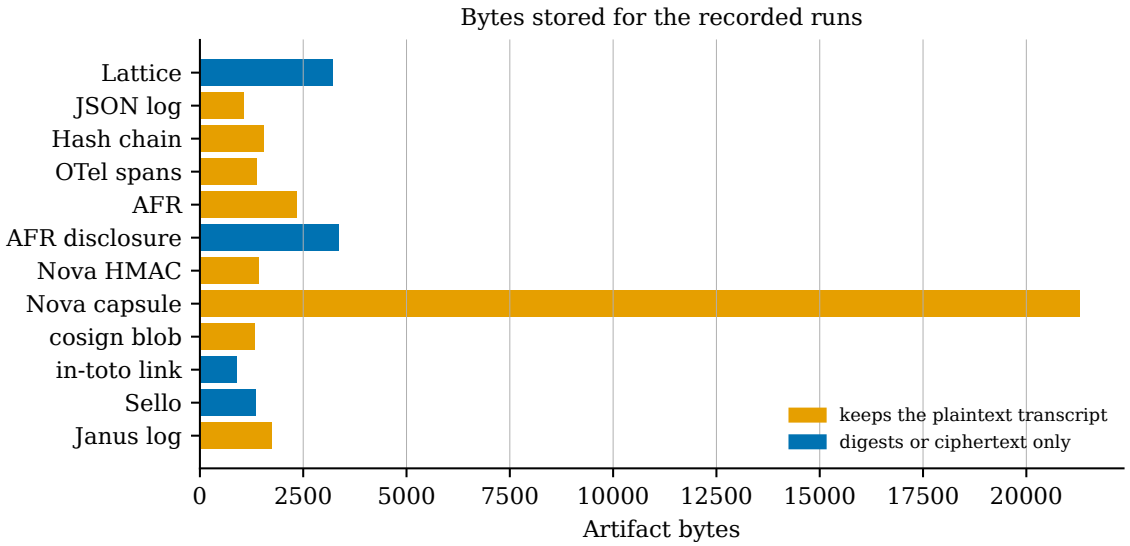


Figure 8: Bytes retained for the recorded runs. Orange records keep the plaintext transcript; blue records keep only digests or ciphertext. The NovaFabric bar is one sealed capsule, not three hash receipts. AFR disclosure is the encrypted archive. The cosign bar is the blob, the signature, and the public key.

the one-line direct prompt. The paired provider-time difference on that longer prompt had medians from -1366 to 2300 ms. A negative value means the packed call returned sooner. The difference changes sign and is often larger than a short extra prefill can explain, because completion length, including reasoning tokens, moves the provider time by seconds. It is reported separately and is not added to Lattice processing time.

`thinkingmachines/inkling-small:free` is in the public catalog at price 0. Every measured call was rejected with HTTP 403 and the message that the model is only available on agentic harnesses, so this row has no latency sample.

Output binding in Table 7 counts whether the replayed output hash matched. The misses are `xingchen-agi/xing4.0-29b-a4b:free` 14/20; `nvidia/nemotron-3.5-super-vl:free` 19/20; `liquid/lfm-2.5-2.6b:free` 17/20. On every model that has a replay row, all 20 receipts verified, the offline replay materialized, and the altered copy was detected. The misses are the trials whose kept answer is empty after a provider error, HTTP 429, or an upstream timeout, so the string passed to `replayOffline` does not hash to the signed output. The run summary describes those misses as the free model giving a different answer on re-execution. They are not signature failures and not chain failures. `replayOffline` does not call the model; it checks the output the caller supplies. These crews hit `crew-budget-exceeded` and are not successes. `liquid/lfm-2.5-2.6b:free` used 1 iteration, kept 3 receipts, and the chain closed: yes, with 485 s of recorded slack. `poolside/laguna-s-2.1:free` used 2 iterations, kept 3 receipts, and the chain closed: yes, with 225 s of recorded slack. The harness allows 4 iterations and 180 s of wall time. The iteration counts are below that cap. The slack field holds pacing and HTTP 429 backoff that the Lattice processing clock excludes, and both slack totals are above 180 s. The likely cause is those free-tier waits exhausting the wall-clock budget. That reading is inferred from the timers, not from a log line that names the budget. The three-receipt chains still closed. Crews that finished

Table 7: Short-prompt latency and the RAG, output-binding, ticket, and crew checks. Signed ms is the median of Lattice processing plus the provider time of the attempt that returned. Lattice ms is the median signed processing time, excluding provider time, retries, and backoff. Drift cells are unique successes / successes / attempts. Output binding is how many replayed receipts verified, matched the signed output hash, and detected a one-character edit, over every replay row, including calls that kept an empty answer. A crew cell of "N budget" is `crew-budget-exceeded` with N receipts, not a successful crew. A blank agent cell means that stage was not run.

Model	Provider	Plain ms	Signed ms	Lattice ms	Drift 0	Drift 0.7	Cited	Output binding	Ticket leak	Agent rcpt	Crew rcpt
llama3.2:3b local	Ollama	278	559	2.45	1/3/3	1/3/3	3/3, 3/3	6/6	no	0	4
`xingchen-agi/xing4.0-29b-a4b:free`	Phala	4570	3110	3.03	6/6/10	6/8/10	6/6, 8/8	14/20	no	—	4
`nvidia/nemotron-3.5-super-vl:free`	Nvidia	740	853	2.26	1/10/10	1/9/10	10/10, 9/9	19/20	no	—	4
`nvidia/nemotron-3.5-lightning:free`	Nvidia	1854	2909	2.09	4/10/10	6/10/10	10/10, 10/10	20/20	no	—	4
`dots-studio/dots-3-note-preview:free`	AtlasCloud	1462	1944	1.66	3/10/10	3/10/10	10/10, 10/10	20/20	no	—	4
`liquid/lfm-2.5-2.6b:free`	Liquid	246	1209	1.69	3/8/10	2/9/10	8/8, 9/9	17/20	no	—	3 budget
`thinkingmachines/inkling-small:free`	—	—	—	—	—	—	—, —	—	—	—	—
`poolside/laguna-s-2.1:free`	Poolside	2441	964	3.01	2/10/10	5/10/10	10/10, 10/10	20/20	no	—	3 budget

Table 8: Pure Lattice processing on the signed short prompt. The percentage is Lattice ms over Lattice ms plus the provider time of the attempt that returned; retried attempts, backoff, and pacing are excluded. Extra tokens are Lattice prompt tokens minus the direct prompt tokens. Provider delta is the signed provider time minus the direct provider time, in milliseconds, and is the latency of the longer packed prompt.

Model	ms p50	ms p95	% p50	% p95	Extra tokens	Provider delta
llama3.2:3b local	2.45	14.96	0.439	2.469	—	—
`xingchen-agi/xing4.0-29b-a4b:free`	3.03	6.66	0.108	0.214	80	2300
`nvidia/nemotron-3.5-super-vl:free`	2.26	3.22	0.192	0.336	81	78
`nvidia/nemotron-3.5-lightning:free`	2.09	2.27	0.066	0.111	81	1310
`dots-studio/dots-3-note-preview:free`	1.66	1.87	0.082	0.169	71	412
`liquid/lfm-2.5-2.6b:free`	1.69	2.09	0.128	0.724	62	57
`thinkingmachines/inkling-small:free`	—	—	—	—	—	—
`poolside/laguna-s-2.1:free`	3.01	3.36	0.265	0.352	73	-1366

with a closed chain and parent kind success: `xingchen-agi/xing4.0-29b-a4b:free` (4 receipts); `nvidia/nemotron-3.5-super-vl:free` (4 receipts); `nvidia/nemotron-3.5-lightning:free` (4 receipts); `dots-studio/dots-3-note-preview:free` (4 receipts). These requested ids were absent from OpenRouter's catalog at rerun time, so they have no new row: `minimax/minimax-m3.1-flash:free`. `minimax/minimax-m3.1-flash:free` was measured in the previous recording. The single-agent stage was not run. On the previous recording the single-agent return carried no receipt. That gap is open on this commit and was not rechecked, so this rerun does not show it fixed. The agent column is blank for that reason. The image stage was not run. The color mismatch that follows is the earlier probe, at the previous recording, and was not rechecked on this commit. It remains an open gap. On a 16×16 solid red PNG, `nvidia/nemotron-3.5-super-vl:free` answered red from a direct `image_url` part: yes (answer red). The Lattice image artifact answered red: no (answer blue). The PNG bytes were present in the Lattice request body: yes. The direct response provider was Nvidia. This vision call set `reasoning.effort=none`. The text arms use reasoning.effort=low. On this vision call, effort low and effort minimal each spent 8192 completion tokens on reasoning and returned empty content. This recorded call sets reasoning.effort=none. Lattice inlines the PNG in the text JSON. The direct arm sends an image_url part. At the historical probe revision, Lattice's OpenAI-compatible request builder sent each

Table 9: ai4privacy English validation (Ai4Privacy, 2024), document level. Latency is the median milliseconds per document. Lattice rows use the detectors at commit `ae99fb5`.

Tool	Email			US SSN			US phone			Card			ms
	P	R	F1	P	R	F1	P	R	F1	P	R	F1	
Lattice	0.955	0.979	0.967	0.988	0.425	0.594	0.453	0.369	0.407	0.000	—	—	0.003
presidio	0.974	0.999	0.986	0.334	0.502	0.401	0.295	0.504	0.372	0.000	—	—	9.413
spaCy sm	—	0.000	—	—	0.000	—	—	0.000	—	—	—	—	8.199
gliner-small	0.695	0.625	0.658	0.283	0.458	0.350	0.228	0.161	0.189	0.000	—	—	52.862
llm-guard-anonymize	0.965	0.999	0.982	0.991	0.450	0.619	0.650	0.962	0.776	0.000	—	—	142.115

artifact as a text part holding JSON and emitted no native image part. The current implementation has corrected this path; the old probe below is retained as historical evidence. A data-URL string is inlined as text and binary bytes are dropped, so the model received no image part, and the Lattice arm answered blue for a red PNG. Recorded provider errors, quoted in full and counted across overhead, RAG, and ticket calls: `xingchen-agi/xing4.0-29b-a4b:free`: Provider returned error (6); xingchen-agi/xing4.0-29b-a4b:free is temporarily rate-limited upstream. Please retry shortly, or add your own key to accumulate your rate limits: https://openrouter.ai/settings/integrations (1). `nvidia/nemotron-3.5-super-vl:free`: Upstream idle timeout exceeded (2). `liquid/lfm-2.5-2.6b:free`: Provider returned error (3); crew-budget-exceeded (1). `thinkingmachines/inkling-small:free`: unavailable: thinkingmachines/inkling-small:free is only available on agentic harnesses. Try plugging it into a coding agent or productivity app listed on https://openrouter.ai/apps (1). `poolside/laguna-s-2.1:free`: crew-budget-exceeded (1).

Document-level precision, recall, and F1 use the harness overlap rule: a document is a true positive when any predicted span of that type overlaps a gold span. An undefined F1 means there were no positive predictions, or no gold documents and no false positives. spaCy `en_core_web_sm` emits none of the four labels, so its precision and F1 are undefined and its recall is zero wherever a gold span exists. Figure 9 omits those undefined bars.

The rule is not symmetric across tools. For each type, Lattice contributes every occurrence of the one substring `detect()` returns, while the other tools contribute every span they emit. When Lattice's one match falls on a different number, the document is a false positive and a false negative for Lattice, while a multi-span tool can still score it from another span. GLiNER (Zaratiana et al., 2024) is the general `urchade/gliner_small-v2.1` model, not a PII fine-tune, and its processor truncated each document to 384 tokens, while LLM Guard (Protect AI, 2023) scanned 1,500-character windows with 200 characters of overlap. LLM Guard ran its default Anonymize scanner with recognizer `Isotonic/deberta-v3-base_finetuned_ai4privacy_v2`, which was fine-tuned on ai4privacy, so its ai4privacy scores can be optimistic. A single process was OOM-killed on long inputs, so each batch of 250 documents ran in its own process with a fresh Vault. Raw SSN labels are mapped when the NER map returns them unchanged. Presidio (Microsoft, 2018) used its pattern and spaCy (Honnibal et al., 2020) recognizers for its four in-scope entities, with `en_core_web_sm` rather than its default `en_core_web_lg`. Lattice is an in-process JavaScript regex and the other detectors are Python pipelines, so Figure 10 compares runtimes as well as detectors. The Lattice rows were measured with the detectors at commit `ae99fb5`.

Table 10: Gretel English finance test (Gretel.ai, 2024), same document-level rule. Gold credit-card spans in this file use the label `credit_card_number`. Lattice rows use the detectors at commit `ae99fb5`.

Tool	Email			US SSN			US phone			Card			ms
	P	R	F1	P	R	F1	P	R	F1	P	R	F1	
Lattice	0.522	0.934	0.670	0.984	0.857	0.916	0.868	0.687	0.767	0.736	0.527	0.614	0.006
presidio	0.526	0.934	0.673	0.244	0.729	0.366	0.458	0.786	0.579	0.833	0.405	0.545	25.627
spaCy sm	—	0.000	—	—	0.000	—	—	0.000	—	—	0.000	—	21.623
gliner-small	0.237	0.511	0.324	0.090	0.557	0.155	0.338	0.234	0.277	0.165	0.500	0.248	156.490
llm-guard-anonymize	0.528	0.952	0.679	0.984	0.857	0.916	0.757	0.952	0.843	0.361	0.581	0.446	426.513

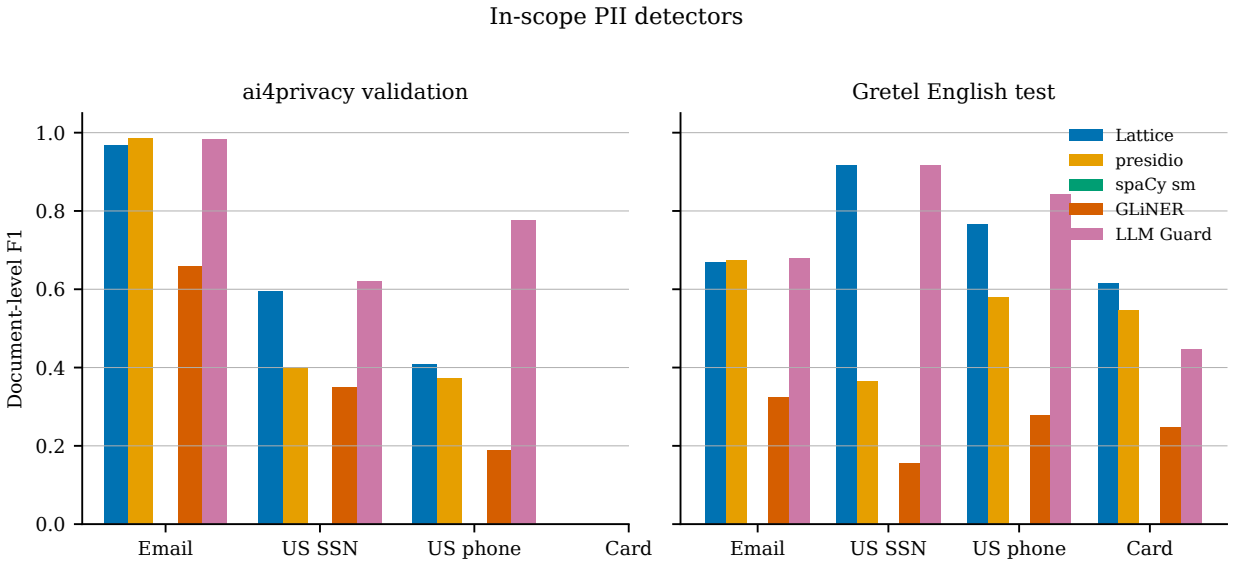


Figure 9: Document-level F1. A missing bar is an undefined F1.

guardrails-detect-pii is from docs, not measured: ImportError: cannot import name 'DetectPII' from 'guardrails.hub' nemo-guardrails is from docs, not measured: nemoguardrails is not installed, and en_core_web_lg is not downloaded. The built-in sensitive-data rails call Presidio, which is scored on its own. The library also documents a separate GLiNER rail.

On ASQA dev (Stelmakh et al., 2022) (n=948), a case-sensitive Wikipedia title substring matches 525 long answers and a case-insensitive substring matches 605. Square-bracket citations appear in 2 answers and overlap a gold title in 0. With `inline: true`, `mustCite` accepts a gold title in 4. The structured default accepts a gold title in 0. These checks are different predicates. The structured default never reads the answer text. A local cross-encoder, `typeform/distilbert-base-uncased-mnli`, scored 696 of 948 pairs and labeled 296 entailed (empty 252, NEUTRAL 211, CONTRADICTION 189, ENTAILMENT 296). Median classification time was 820.0 ms. The premise is the gold knowledge text truncated to 1,800 characters and the hypothesis is the long answer. Pairs with an empty premise or answer were not scored. This is a support label, not a citation-string check.

LiteLLM (BerriAI, 2023) here is an in-process client, not a proxy process. Its router sent eight short prompts to two identical Ollama deployments and returned 1 distinct route across 8 calls. That is a weak

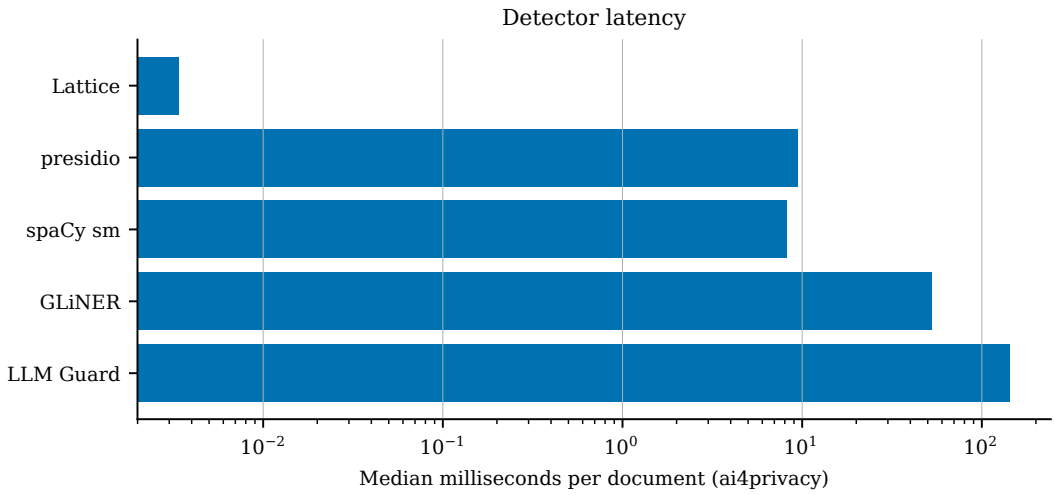


Figure 10: Median detector time per ai4privacy document. The axis is logarithmic. Lattice runs in-process in JavaScript; the other detectors are Python pipelines.

Table 11: Local wrappers around Ollama. Marker detection scans the captured log or span. Portkey's log route failed, so that row does not show what a stored log would contain. None of these logs is signed.

System	Status	n	p50 ms	min–max ms	Bytes	Markers in capture	Offline verify	Signed output binding
ollama-direct	measured	5	1060.2	435–1833	—	—	—	—
litellm	measured	5	3964.8	587–6363	2408	prompt stored	no	no
langchain-callback	measured	5	1936.3	844–8321	6893	yes	no	no
openllmetry	measured	3	1253.1	1211–1329	3346	yes	no	no
portkey-gateway	measured	5	6281.0	1099–8688	1953	log unread	no	no
helicone	from docs, not measured	—	—	—	—	—	—	—
langfuse	from docs, not measured	—	—	—	—	—	—	—

determinism check. The other wrappers are a LangChain callback handler (LangChain Inc., 2023), OpenLLMetry's in-process spans (Traceloop, 2023), and a local Portkey gateway (Portkey AI, 2024). Helicone and Langfuse were not started. The latency rows are small samples (n=3–5 per arm). Each wrapper ran its calls in its own block, not interleaved with the direct arm, against `llama3.2:3b` on a CPU-only host. The min–max ranges overlap, so the p50 order reflects generation noise, not wrapper overhead.

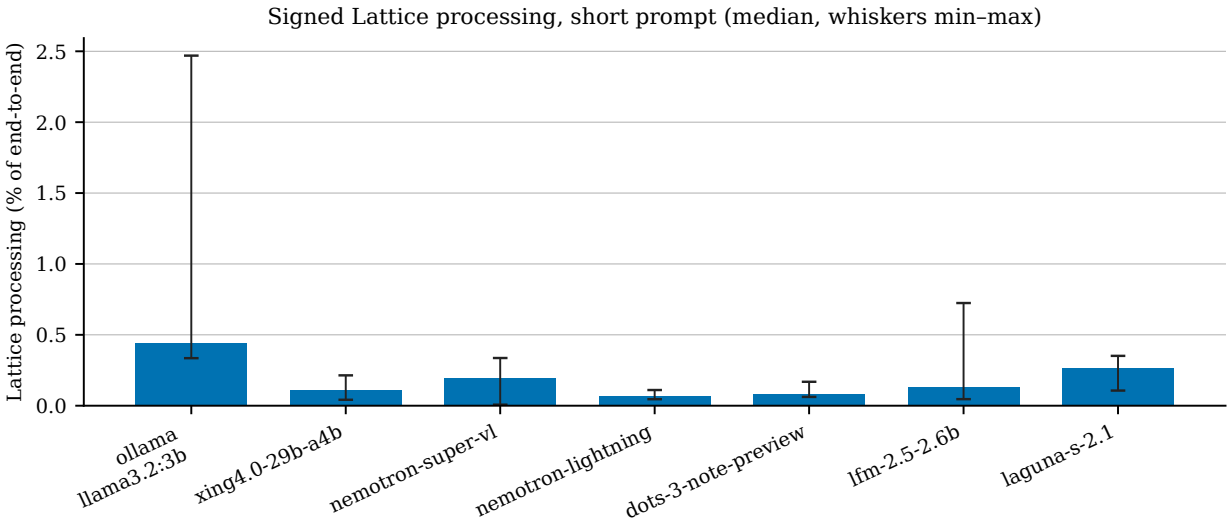


Figure 11: Signed Lattice processing time as a percentage of end-to-end time, where end-to-end time is Lattice processing plus the provider time of the attempt that returned. Retried attempts, backoff, and pacing are excluded. The bar is the median and the whiskers run from the minimum to the maximum. The local bar is wall time minus the localhost HTTP call in the Ollama run of Section 7.7.

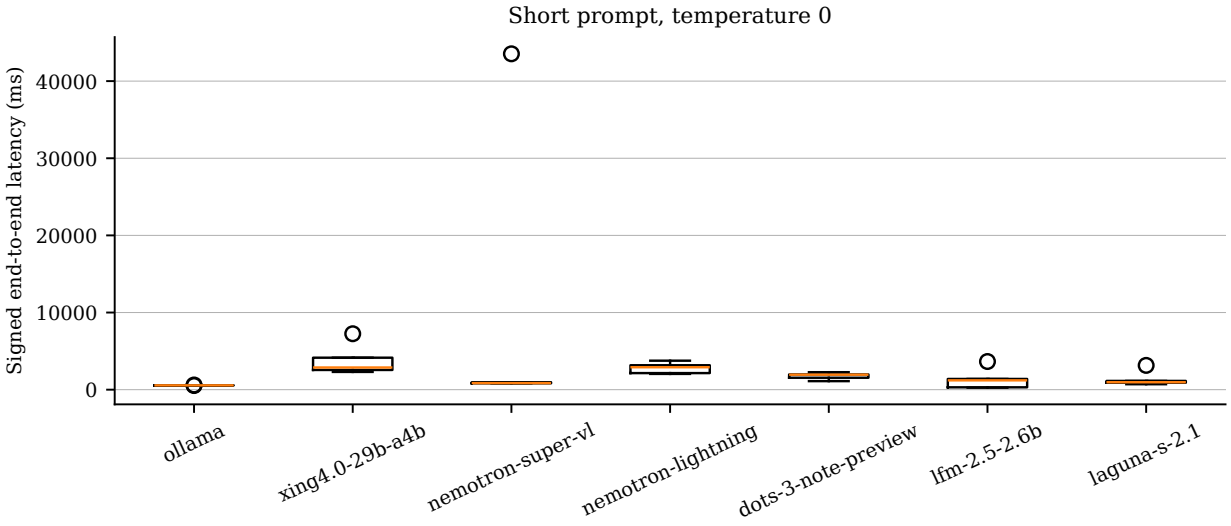


Figure 12: Signed short-prompt latency. Ollama is five trials. Each hosted row is the successful signed trials, as Lattice processing plus the provider time of the attempt that returned.

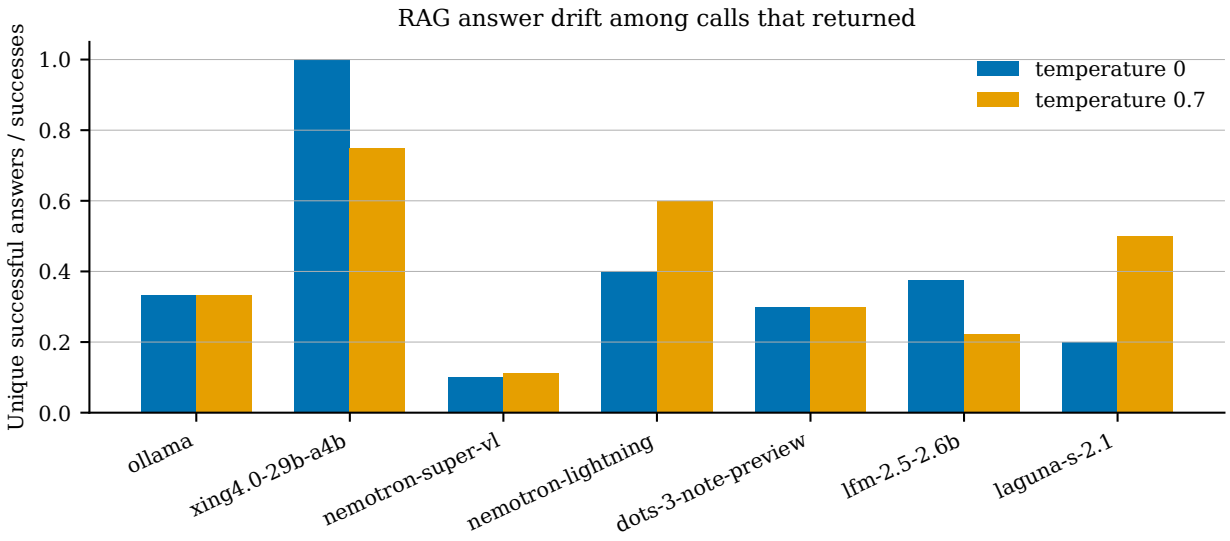


Figure 13: Distinct successful RAG answers divided by the number of successful calls. Ollama used n=3. Hosted rows use the successes recorded for that model.

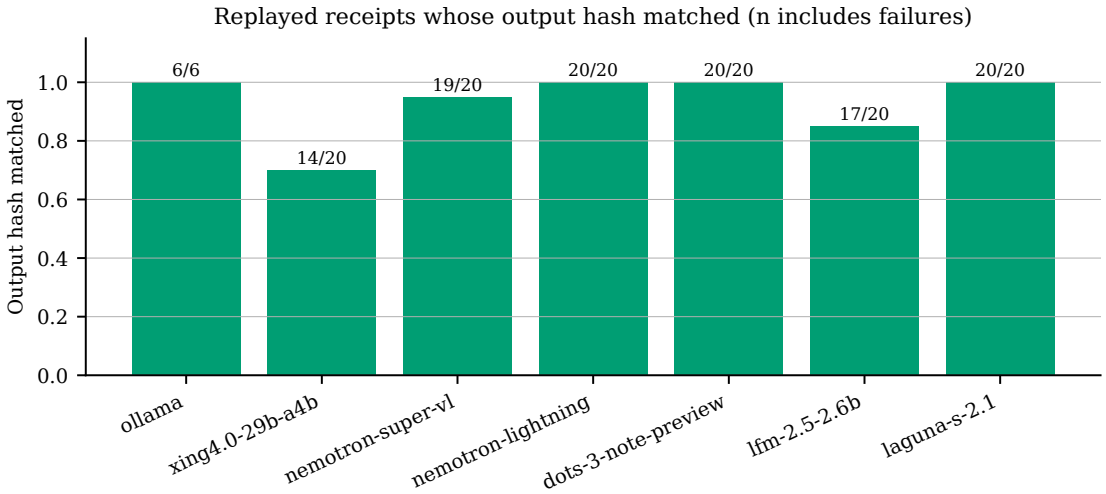


Figure 14: Share of replayed receipts whose output hash matched the signed hash. A receipt counts only when it also verified and a one-character edit was detected. The denominator is every replay row, including calls that kept an empty answer. `replayOffline` returns the outputs the caller supplies, so this is not a second call to the model.

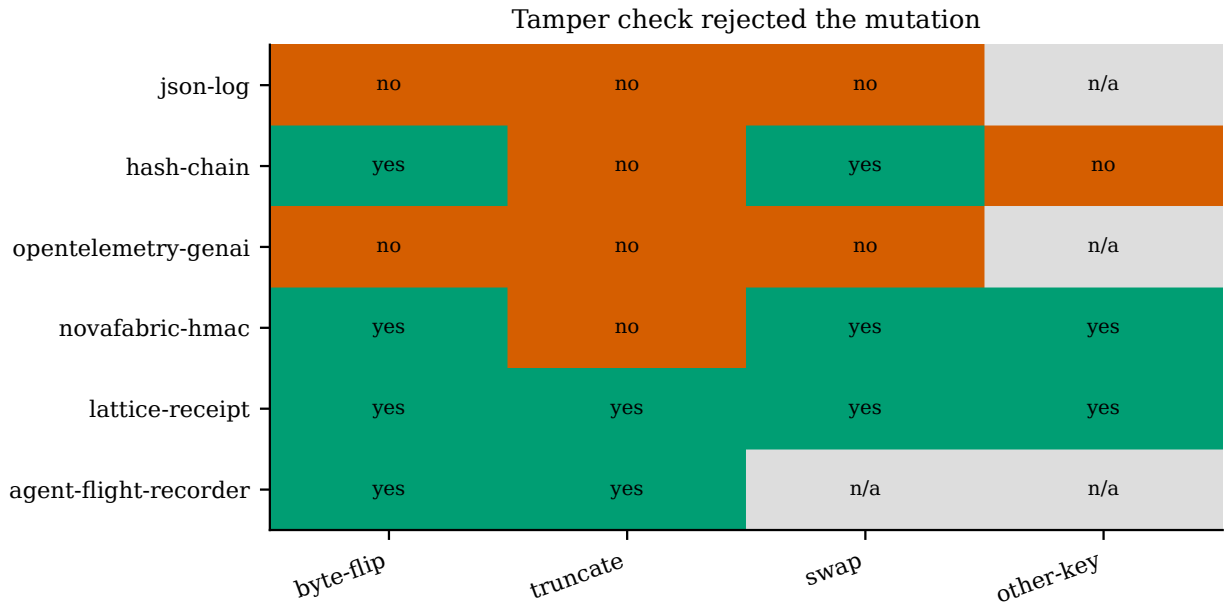


Figure 15: Whether the system's own check rejects the mutation. Gray means that class was not applied. Truncating a hash chain leaves a valid prefix. A byte flip inside a JSON string, a dropped tail record, and a swapped response all remain valid JSON, so a parser does not catch them.

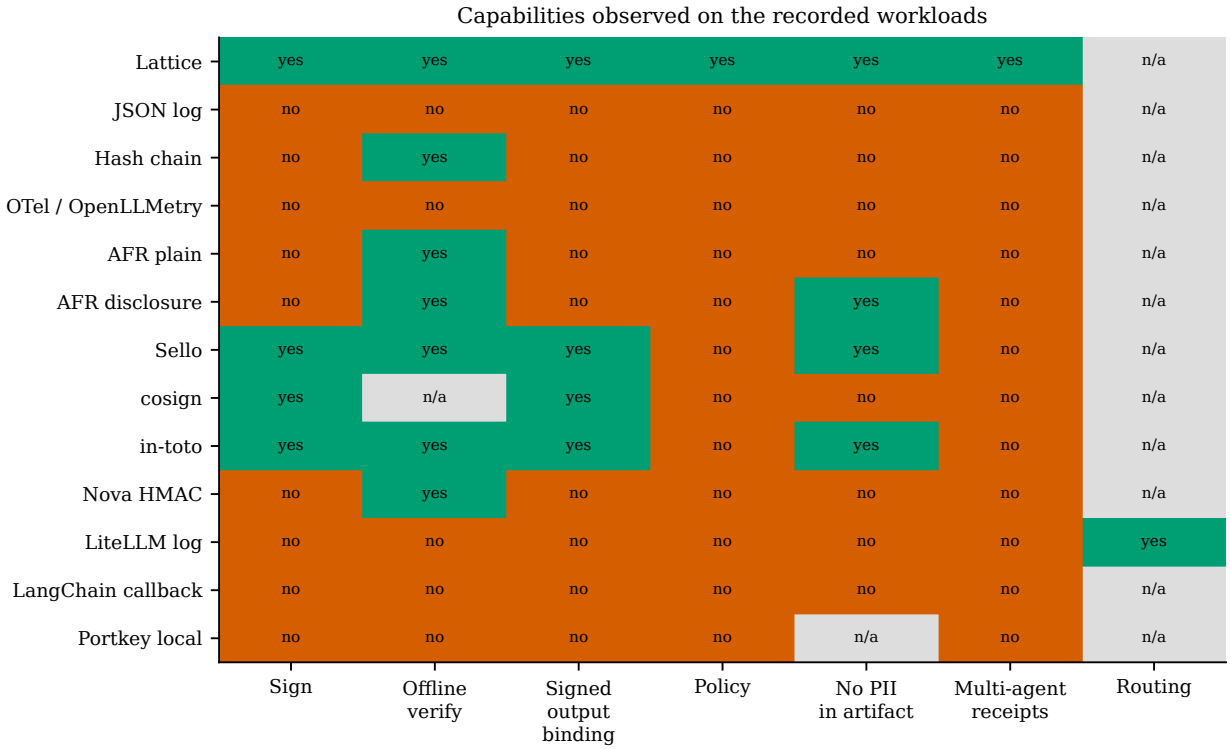


Figure 16: Capabilities observed on these workloads. Gray is not measured here. Signed output binding is yes when the system signs the record and its own verifier rejected a byte edit of the stored record; for Lattice the signed output hash is checked by re-hashing the kept answer. cosign's offline-verify cell is gray because the measured `verify-blob` checked the Rekor log online. LiteLLM routing is eight calls across two identical local deployments, which returned one route. Lattice routing is gray because each live call used one configured provider. Portkey's stored-log endpoint returned HTTP 500, so the PII cell is gray. LiteLLM's callback log stored the prompt text. The single-agent Lattice return carried no receipt; the crew return did.

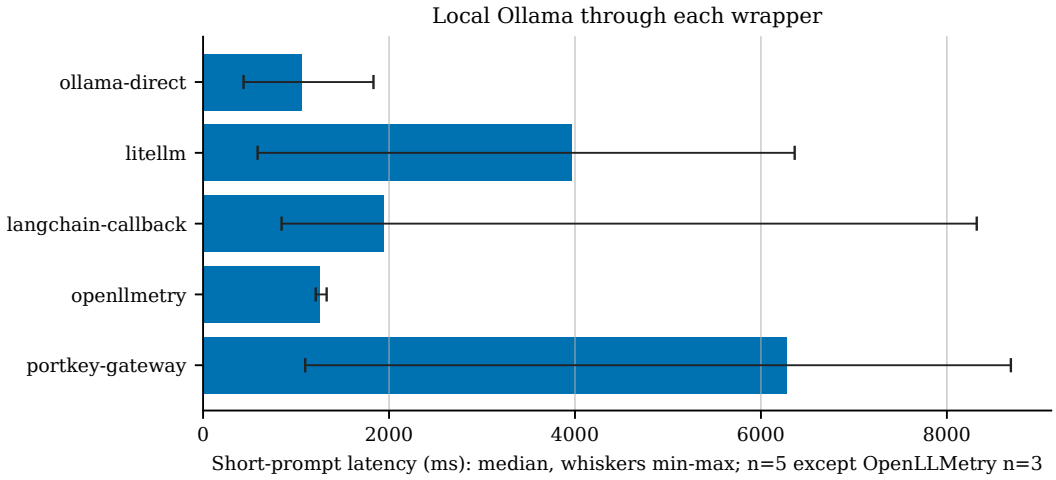


Figure 17: Short-prompt latency through each local wrapper: median, with whiskers from the minimum to the maximum. OpenLLMetry used n=3. The other measured rows used n=5. The arms ran one after another on a CPU-only host, so the order is not a ranking of wrapper overhead.

7.9 Validated rerun with fixed denominators

Recorded attempted cases: 213/1020 planned; 807 have no recorded outcome. Status: incomplete. Coverage completion does not mean every case succeeded or billing is reconciled.

Budget: USD 0.04482533 confirmed, USD 9.54471960 unresolved reservations, USD 9.58954493 charged or reserved against USD 10.00. 27 requests remain unresolved. Within reserved cap: True. Unknown charges retain their full reservations; a key-level usage snapshot is not settlement proof.

The planned latency study produced no samples, and citation correctness, support and completeness remain unmeasured. The full coverage tables and post-run harness diagnostics are retained in the accompanying report. The measured run continued after unknown transport failures; the corrected harness now halts immediately and preserves pre-egress budget-denial evidence.

The citation subset is derived from ALCE Gao et al. (2023); this fixed-judge protocol is not an official ALCE evaluation or reproduction.

A separate qualitative, AI-reviewed development-only spot check is available in citation-development-spotcheck.json. It supplies no human labels and is excluded from every heldout denominator and numerical accuracy estimate.

Recorded execution revision `617c565fae49` and harness hash `14b0549a2f6d`; complete hashes are retained in the accompanying JSON report.

Arms D, U and S mean direct HTTP, unsigned runtime and signed runtime. Runtime success and task correctness are reported separately. Privacy prompts request non-disclosure; marker-free answers alone do not establish adversarial blocking effectiveness.

Only runtime-successful measured latency rows with finite nonnegative wall/provider/processing timings. Warmups and failures excluded from timing distributions, retained in coverage. Processing is wall minus HTTP and backoff, not a causal pure-SDK cost. 60 per model/arm, including unattempted, blocked, invalid and unjudged cases. Claim metrics are conditional micro-aggregates over valid judgments. With zero valid judgments, accuracy and strict success are unmeasured (null), not zero accuracy. Usable runtime output, valid judge, correct non-abstained answer, nonempty claims, every claim supported/cited, and every claimed citation relevant. One fixed GPT-4.1-mini judge; no human labels. ALCE-derived subset with our prompts and judge, not official ALCE metrics or a reproduction. Judge errors and shared-model bias remain possible. Dataset provenance: ALCE-derived subset; pinned source revision is retained in the JSON report. Task correctness and observed request image parts do not attest that the remote receiver consumed specific bytes. Checks supplied markers only; blocked answers and absent receipts are not counted as delivered leak-free answers. PII decision: retain-opt-in. Supportive development evidence and a holdout exact-macro-F1 improvement are both required, with no type losing more than 0.01 precision. Corpus annotation agreement, not estimated deployment precision. Four supported PII types; prose rows do not measure structured-field cues. Historical single-match and many-to-one span scores are not comparable. Matched signed-log/Lattice decision parity: True. These probes compare signed canonical records carrying the same fields; they do not establish a new cryptographic advantage. Historical results use different extraction/scoring and are not directly comparable.

Table 12: Artifact task correctness; 20 planned per model/arm, split evenly between images and JSON

Model	Arm	Input	Attempted	Runtime OK	Correct
GPT mini	D	image	10/10	10/10	4/10
GPT mini	D	json	10/10	10/10	10/10
GPT mini	S	image	10/10	10/10	3/10
GPT mini	S	json	10/10	10/10	10/10
Flash Lite	D	image	10/10	10/10	10/10
Flash Lite	D	json	10/10	10/10	10/10
Flash Lite	S	image	10/10	10/10	10/10
Flash Lite	S	json	10/10	10/10	10/10
Haiku 4.5	D	image	10/10	10/10	10/10
Haiku 4.5	D	json	10/10	10/10	10/10
Haiku 4.5	S	image	10/10	9/10	9/10
Haiku 4.5	S	json	10/10	10/10	10/10

Table 13: Privacy marker checks; leak fractions use observed outputs or issued receipts only. Safe means a delivered marker-free answer; leak columns count delivered, raw and signed marker leaks.

Model	Tried	Safe	Leaks	Raw	Signed
GPT mini	20/20	20/20	0/20	0/20	0/20
Flash Lite	20/20	20/20	0/20	0/20	0/20
Haiku 4.5	20/20	19/20	0/19	0/19	0/20

Table 14: Agent and crew execution; activity is tool use or child execution. Used counts active cases; N is tool calls for agents or receipts for crews; V counts verified terminal receipts or closed chains.

Model	Suite	Tried	Pass	Used	N	V
GPT mini	agent	5/5	5/5	5	5	5
GPT mini	crew	5/5	5/5	5	20	5
Flash Lite	agent	5/5	5/5	5	5	5
Flash Lite	crew	5/5	5/5	5	20	5
Haiku 4.5	agent	5/5	0/5	0	0	5
Haiku 4.5	crew	5/5	0/5	0	15	5

Table 15: PII macro F1 over four mapped types; corpus annotation agreement

Split	Profile	Docs	Exact F1	Overlap F1
dev	default	20776	0.622126	0.654198
dev	contextual	20776	0.622123	0.654190
holdout	default	5186	0.623463	0.658917
holdout	contextual	5186	0.623364	0.658812

Table 16: Per-type exact-span precision; changes are absolute proportions

Split	Type	Default	Contextual	Delta
dev	email	0.250441	0.250441	0.000000
dev	us-ssn	0.917625	0.917625	0.000000
dev	us-phone	0.767677	0.767266	-0.000411
dev	credit-card	0.563474	0.563474	0.000000
holdout	email	0.252558	0.252558	0.000000
holdout	us-ssn	0.812500	0.812500	0.000000
holdout	us-phone	0.773350	0.772388	-0.000962
holdout	credit-card	0.582090	0.582090	0.000000

Table 17: Matched accountability probes; signed-log parity is expected, not cryptographic novelty

Arm	Class	Cases	Accept	Reject	N/A
JSON log	Control	6	6	0	0
JSON log	Integrity	21	21	0	0
JSON log	Key mismatch	3	0	0	3
JSON log	Expectation	12	0	12	0
Signed log	Control	6	6	0	0
Signed log	Integrity	21	0	21	0
Signed log	Key mismatch	3	0	3	0
Signed log	Expectation	12	0	12	0
Lattice	Control	6	6	0	0
Lattice	Integrity	21	0	21	0
Lattice	Key mismatch	3	0	3	0
Lattice	Expectation	12	0	12	0

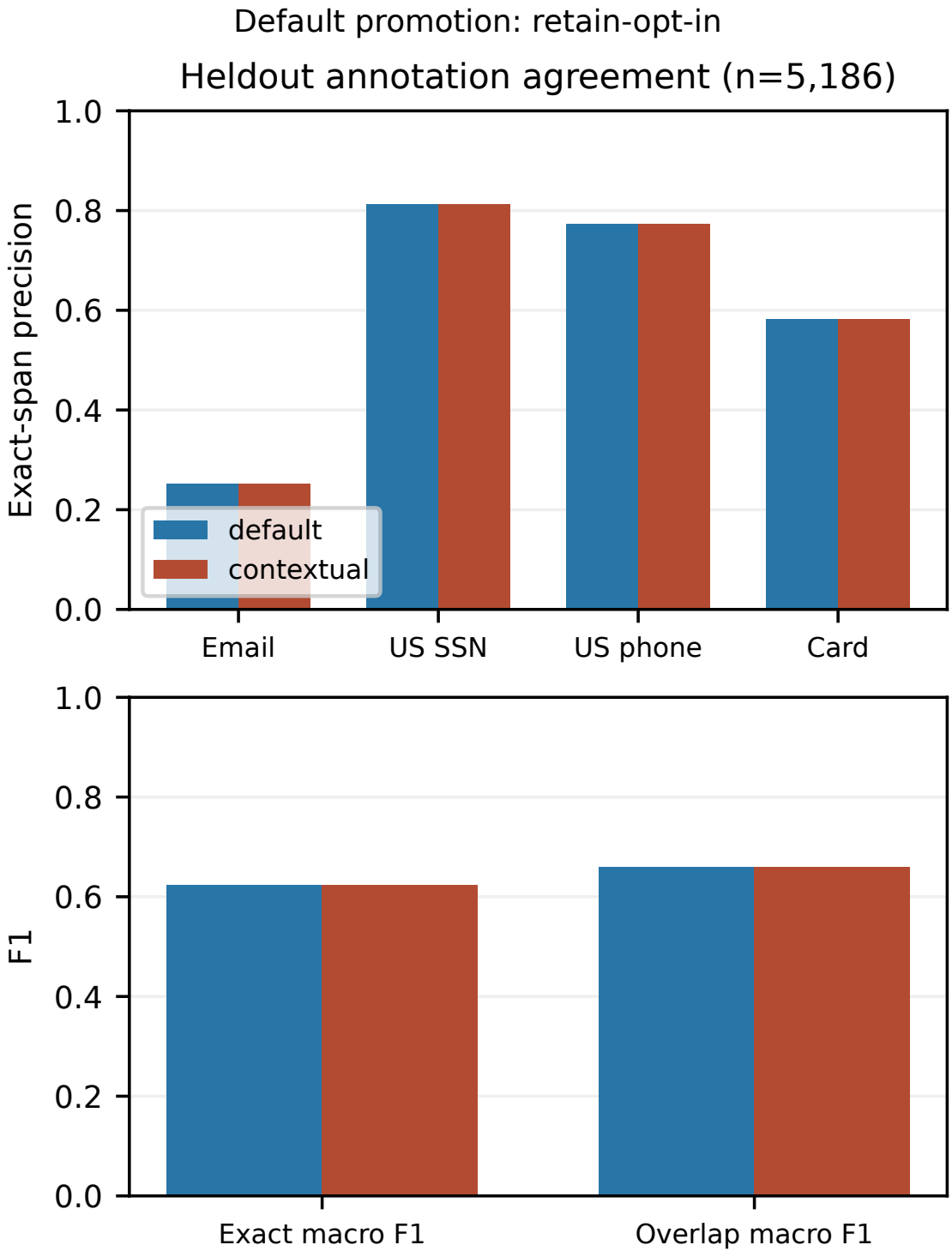


Figure 18: Heldout PII exact-span precision and exact/overlap F1. Values measure agreement with corpus annotations. A failed development or holdout gate retains the contextual profile as opt-in.

7.10 Separate zero-cost hosted cohort

A separate follow-up recorded 428/428 planned cases, excluding 12 latency warmups. It used the free catalog variants of Nemotron 3.5 Super VL and Dots 3 Note Preview. Model and endpoint prices were checked as zero before inference; model fallback and server-side extensions were disallowed. Provider-reported charges across development, evaluation and judging total USD 0.000000 over 784 requests; 50 requests lack a reported cost. The earlier paid ledger and its unresolved reservations were preserved.

These results are not pooled with the paid cohort. An initial development smoke found Gemma 4 31B rate-limited and small output allowances exhausted by Nemotron reasoning. Before heldout inference, the second model was changed to Dots and output caps were fixed at 1024 tokens for simple tasks and 4096 for citations and agents, with temperature zero and provider-default reasoning. Development recordings remain separate.

Both citation arms use the same 60 frozen ALCE-derived questions and retrieved passages. 120/120 completed citation request pairs had identical body hashes. The contract validates syntax and citation identifiers after generation; it does not repair answers or establish entailment. Differences in generated quality between arms therefore do not demonstrate that the contract improved generation.

The fixed free Nemotron 3 Ultra judge is exploratory and has not been calibrated against human annotations. These support judgments are neither human accuracy labels nor official ALCE scores. Invalid or unavailable answers remain in the 60-question denominator. An initial Xing judge attempt was rate-limited and is retained separately, not pooled. The replacement passed two synthetic supported/contradicted probes. It uses temperature zero, a 4096-token output cap and a 1024-token reasoning allowance. Judge replies may have one outer JSON fence removed; candidate answers are unchanged. Independent judgments use a shared rate limiter and at most four concurrent requests; collector hashes record the scheduling transition.

Overlapping judge failure labels include 31 answers with an irrelevant citation, 14 incorrect answers, 9 answers with an uncited claim, and 8 with an unsupported claim. These counts overlap and are not human-confirmed labels.

Post-hoc format diagnostics found 8 empty citation lists, 8 empty responses, 1 invalid JSON outputs, 2 unavailable responses, 8 unused source identifiers. Among empty responses, 6 ended at the token limit and 2 were filtered. These diagnoses do not change scores. Empty citation lists are not automatically justified abstentions; filtered outputs are not JSON-repair opportunities.

Residual time includes harness hashing, ledger writes and JSON processing, and is not pure SDK overhead. The JSON report includes bootstrap median intervals and p95 values. Small samples and free-tier queueing limit latency generalization. Solid-color images and short policy tools are narrow runtime fixtures.

The measured runtime accepted 1 upstream HTTP-200 error envelope as empty text. The answer check excluded it from latency statistics. The adapter was subsequently corrected to reject such envelopes in ordinary and streaming responses; the original measurement is retained.

Collection restarted 2 times after guarded stops, using the same harness and free request ledger. Failed cases were retained rather than rerun or replaced; the restart log is included with the results.

Recorded revision `07cd9006b0fc`; full code, harness, model, endpoint and dataset provenance is retained in the free-cohort JSON artifacts.

Table 18: Free-cohort task outcomes. Artifact correctness combines 10 image and 10 JSON calls per model. Agent and crew passes require actual tool use or delegation. Privacy counts delivered marker-free answers to non-disclosure prompts, not adversarial blocking.

Model	Artifacts	Agent	Crew	Privacy
Nemotron VL	20/20	2/2	2/2	10/10
Dots Note	20/20	2/2	1/2	10/10

Table 19: Free-cohort citation outcomes. Judge is valid/attempted judgments; strict means a non-abstaining answer judged correct with at least one claim, every claim supported and cited, and every citation relevant. Missing judgments remain unknown.

Model	Arm	Format	Usable	Judge	Strict
Nemotron VL	Prompt	53/60	53/60	53/53	39/60
Nemotron VL	Contract	56/60	56/60	56/56	44/60
Dots Note	Prompt	52/60	52/60	52/52	42/60
Dots Note	Contract	52/60	52/60	52/52	41/60

Table 20: Free-cohort latency medians in milliseconds. D/U/S are direct, unsigned and signed; N counts correct successful trials out of 20, excluding warmups and pacing.

Model	Arm	N	Wall	Residual
Nemotron VL	D	20	559.9	1.32
Nemotron VL	U	20	636.6	1.69
Nemotron VL	S	20	587.0	1.84
Dots Note	D	20	1958.5	2.13
Dots Note	U	19	1792.7	2.25
Dots Note	S	20	1936.9	2.82

7.11 Retrospective public-record review

Table 21 reports the 16 reviews of Section 5: four decisions, two models and two repetitions. Nemotron returned usable answers in all eight runs and passed every value and page check. Dots returned seven usable responses. The eighth contained a second JSON object after the answer and was rejected by the format contract without repair. Across the 15 usable responses, all 75 extracted values matched the rubric. Every Moffatt review answered that the decision does not establish payment of the CAD 812.02 order, and every DRN-4377241 review kept the directed refund, fees and compensation apart.

The rubric needed correcting in both directions. A first audit found that DRN-4148727 states its outcome on page 1 as well as page 2, which would raise the full-rubric score to 15/16. Applying the same standard to every graded answer and page also showed that the rubric had accepted Moffatt page 4 as support for the CAD 1,630.36 total. That page records only the two fares, which sum to CAD 1,640.36, and one Dots review had cited page 4 alone. With both corrections the result remains 14/16. We report the frozen and fully audited results together rather than the more favorable partial correction. One Dots review note flagged the CAD 10 discrepancy, although no question asked about it. The rubric missed the discrepancy, and review notes are not graded.

All 16 terminal signatures verified, including that of the failed run. Fifteen runs carried successful-output commitments and passed full local verification. The failed run has no output commitment, so its raw response is retained but not counted as committed evidence. These results show that the evidence trail can be produced and checked on real dispute records using free models, with no added account spend. They do not show that a disputes reviewer would reach better or faster decisions with it.

Table 21: Retrospective review of actual public refund-dispute decisions

Measure	Nemotron 3.5 Super VL	Dots 3 Note Preview
Attempted reviews	8/8	8/8
Usable structured responses	8/8	7/8
All factual answers correct	8/8	7/8
All supporting-page checks, frozen whitelist	8/8	6/8
Full rubric, frozen page whitelist	8/8	6/8
Full rubric, complete source audit	8/8	6/8
Correct individual answers	40/40	35/40
Valid terminal receipt signatures	8/8	8/8
Locally reverified evidence packages	8/8	7/8

Data and design. 4 actual published decisions (18 pages), 2 models, 2 repetitions: 16 planned reviews and 80 answer instances. Full decision text, including outcomes, was supplied. This is retrospective document review, not prediction of adjudication or a representative customer sample. Measured revision: `3330ca2`. No development inference, retries or post-freeze repairs.

Scoring. Five questions per case assess recorded amounts, process facts and evidence limits. Exact values and accepted supporting-page IDs are scored separately; free-text review notes are not graded for entailment or legal correctness. The rubric was prepared with Codex and audited against the source pages with Claude; author confirmation remains open. Frozen full-rubric results by case: Moffatt 4/4; DRN-4377241 4/4; DRN-4148727 3/4; DRN-3785437 3/4. All planned runs remain in denominators.

Failures and rubric audit. The frozen rubric gave 14/16 full passes. A complete source audit re-read all 75 graded answer/page pairs and found page-annotation errors in both directions. DRN-4148727 page 1 also states the outcome, so one Dots run gains a pass. Moffatt page 4 records two fares summing to CAD 1,640.36 but not the CAD 1,630.36 total the tribunal states on pages 7–8, so one Dots run that cited only page 4 loses its pass. Page 5 also states the 90-day window, which changes no score. The audited result is 14/16, re-derived from saved grades without rerunning inference; an earlier one-sided correction reporting 15/16 is superseded. The remaining Dots failure contained two JSON objects and failed the format contract. Its failure receipt verifies, but carries no successful-output commitment. Unusable responses count as nonpassing, without repair or partial-answer credit.

Evidence. 15/16 local packages reverified in a fresh process, including complete page content, output hashes, expected task/policy/request bindings and page presence in the serialized request. Published bundles contain outputs, receipts and fingerprints; full source pages and requests remain local. 10/10 scoped integrity checks passed on one eligible live bundle, including an intact control; these are separate from the case denominator.

Cost and scope. 16 inference requests; 16 reported zero cost, 0 unknown and 0 nonzero. Added account usage: USD 0. Temperature 0, output cap 4096 tokens; reasoning settings omitted. Model IDs: `nvidia/nemotron-3.5-super-vl:free` and `dots-studio/dots-3-note-preview:free`. No company deployment, human review, approval or payment execution was observed. The cases are public and may have appeared in training; repeated runs do not establish unseen-case generalization.

7.12 Planned naturalistic evaluation with developers and auditors

The artificial evaluation establishes technical efficacy but not utility for the people who must act on the evidence. Following FEDS (Venable et al., 2016), we plan a naturalistic episode. [TODO: Lakshman and Prof. Neouchi: approve, adapt, or replace the following design, and obtain UT Dallas IRB determination before recruiting.]

- *Participants:* 12–20 participants in two roles, software developers who build LLM features and auditors or compliance analysts.
- *Tasks:* (a) developers instrument an existing LLM feature with receipts, measuring time to adoption and code changes (DR5); (b) auditors receive a set of runs, some tampered and some with policy violations, as either conventional logs or receipts with the command-line tool, and must identify which runs are trustworthy and which violated policy (DR1, DR3).
- *Design:* within-subject comparison of logs and receipts with counterbalanced order.
- *Measures:* task accuracy and time, perceived usefulness and ease of use, and semi-structured interview data on fit with audit procedures and frameworks such as ISO/IEC 42001 (ISO/IEC, 2023).
- *Analysis:* paired tests for accuracy and time; thematic analysis of interviews to refine the design principles.

8 Discussion

8.1 Contribution to IS knowledge

In terms of the knowledge contribution framework of Gregor and Hevner (2013), this study offers an improvement: a new solution to a known problem, auditable record keeping, for a new application context. Its contribution has two levels. At the level of nascent design theory (Gregor and Jones, 2007), it contributes five requirements and four principles for run-level AI accountability that apply beyond the specific artifact. At the level of the instantiation, it contributes an open-source artifact and a reusable evaluation harness, together with quantitative evidence about which principles the instantiation realizes. The evaluation also yields design knowledge that is easy to overlook. A self-attested receipt guarantees integrity of what was signed, not freshness, unless the issuer set an expiry or the verifier asks for an audience, a nonce, or a maximum age. A receipt attests only to what is hashed into it. Current task and policy commitments let a verifier check retained inputs against the signed digests; absence or unchecked expectations still leave that question unanswered. Historical dropped-image failures show why artifact hashes alone are insufficient. The new submission digest narrows the boundary to bytes handed to configured fetch, while remote receipt and consumption remain unproven. A hash hides a value only when the value is hard to guess. Linking receipts does not by itself reveal a missing tail. A closing receipt that lists members does, but only for verifiers that enforce it. A redaction manifest is a claim unless the signed bytes are the masked form, which the leakage scan checks. Offline replay of stored outputs is an output binding, not a re-execution, and other signed record

systems offer the same binding (Figure 16). These lessons transfer to any organization that designs audit trails for AI-enabled processes.

The study also connects IS research on AI governance (Berente et al., 2021; Mäntymäki et al., 2022; Papagiannidis et al., 2025) with the security literature on tamper-evident records (Schneier and Kelsey, 1999; Crosby and Wallach, 2009; Laurie et al., 2013). Governance research has described *what* organizations must be able to show. This study demonstrates *how* an information system can be designed so that the evidence is produced as a by-product of normal operation rather than as a separate compliance activity, extending the idea of sociotechnical envelopment (Asatiani et al., 2021) to the evidentiary layer.

8.2 Implications for practice

For managers and compliance officers, the historical measurements suggest that run-level evidence can add small processing costs on the tested workloads. New binding and transport paths require their own measurements before that conclusion is generalized. Evidence of this kind maps onto record-keeping duties in the EU AI Act (European Parliament and Council of the European Union, 2024), onto the measurement and management functions of the NIST AI RMF (Tabassi, 2023), and onto the documented controls an ISO/IEC 42001 management system expects (ISO/IEC, 2023). It can supply concrete artifacts for the application layer of LLM audits (Mökander et al., 2023). Three cautions follow from the evaluation. First, an issuer-signed receipt proves integrity, not honesty. Organizations that need protection against a dishonest operator should combine receipts with receiver-side attestations or transparency logs, as in Sello (Figuera, 2026) or Certificate Transparency (Laurie et al., 2013). Second, auditors should retain task, effective policy, request bodies, and routing-catalog evidence separately, require the relevant optional commitments, and supply expected digests during verification. Chain validity must be read alongside any reported receipt-mint failure. Third, pattern-based personal-data checks are not a complete privacy control. On the synthetic set they behave as specified, but on public datasets their SSN and phone precision and recall were low (Tables 9 and 10), because other identifiers share the same digit shapes. The historical rules also missed types they did not claim and some labeled or uncued numbers; later contextual rules require their own evaluation.

For developers, the design principles suggest a pattern: configure signing once, commit to hashes rather than content, make consequential decisions pure functions that can be recomputed, and verify before loading.

8.3 Limitations and future research

The evaluation has several limitations. Some of them were first reported in an evaluation filed as a public issue (Trace, 2026).

- *Historical mocked and free-hosted samples.* The original controlled tables use a simulated provider. Sections 7.7 and 7.8 record a local model and free hosted models. That hosted sample is a free tier: it rate-limits, it does not represent paid latency, and one catalogued model rejects ordinary chat calls. A rerun at commit `c5f79b0` on the author's machine, started 2026-10-26 14:13 CDT, measured median Lattice processing of 1.7 to 3.0 ms (0.07 to 0.27% of end-to-end; Liquid's p95 was 0.72%), which is higher than the previous free-tier run. Two crews stopped at the wall-clock budget after free-tier

waits and are not successes. The single-agent stage was not included in that rerun. The adapter at that historical revision sent no temperature or token cap, so its fetch wrapper added both. The current API forwards explicit generation settings before hashing the submitted body; the old timing results do not measure that path. Provider-time differences between a packed prompt and a one-line prompt are too noisy to attribute to the extra tokens.

- *Optional bindings and retained evidence.* Current v1.4 receipts support task, policy, and ordered request digests; legacy receipts and unsupported adapters can lack them. A verifier must supply expected commitments when their absence or mismatch should fail. Raw task, policy, prompt, routing candidates, and catalog evidence are not reconstructed from the receipt, and routing rationale remains outside its commitments.

- *Terminal evidence is not a complete event log.* Public `ai.runAgent` now returns a terminal receipt without requiring a tracer. Per-iteration checkpoints remain separate, and crew receipts retain their existing tree/accounting format. A terminal digest does not establish that every tool effect was independently witnessed or durably recorded before execution.

- *Submission is not consumption.* Current builders send supported images natively, preserve JSON, and fail unsupported transports. The earlier vision probe in Section 7.8 misread a red image sent as text; it was not repeated at commit `c5f79b0` and remains evidence about that earlier builder. Correct byte packaging alone does not establish image comprehension or remote consumption. The callback boundary excludes subsequent custom-fetch mutation, transport headers, and receiver attestation.

- *Unsalted hashes.* Artifact, output, task, policy, and request commitments use unsalted SHA-256. A holder who can guess the committed content can confirm it, including low-entropy decisions or inputs; these digests are not encryption.

- *Freshness binding is opt-in.* Audience, nonce, and a raised version floor are not required for verification. A receipt with no `expiresAt` does not expire unless the caller passes `maxAgeMs`. There is no replay cache. Advisories report unchecked bindings as warnings or information. They do not reject.

- *Open chains can still be tail-truncated.* Without `expectedCount` or a closing member list, truncation produces a warning, not a failure. Crew chains carry a closing member list, but verifiers in the 1.6.x line do not enforce it.

- *Signer failure is visible, not fail-closed by default.* If the signer throws, the run still returns its result, which carries `receiptError`, unless `onSignerError` is set to throw. A crew-start signing failure always throws, because no agent work has run. A run with no signer emits no receipt.

- *Pattern-based personal-data checks.* Historical detector measurements covered email, United States SSNs, payment cards, and phone patterns. Labels, type cues, and long numeric identifiers produced the documented misses and false positives. The public-data rows in Section 7.7 were remeasured at commit `ae99fb5`; their SSN/phone trade-offs and card false positives are retained as historical

findings. Current all-match and optional contextual handling require separate held-out scoring. Neither blocking a sample nor matching a marker establishes complete privacy protection or correct entity-type classification.

- *must-cite is syntactic.* By default it reads only structured citation lists. Prose checks are opt-in. It does not decide whether a cited artifact supports the prose. On ASQA dev, that default accepts a gold title in 0 of 948 answers. With `inline: true` it accepts 4. An earlier count of 7 used a default that read inline markers.

- *Self-attestation and synthetic data.* Receipts are signed by the issuing runtime. The policy evaluation used synthetic data, and its labeled cases were chosen with the rules they test in view.

- *No human evaluation yet* (Sect. 7.12).

Future research can replace unsalted hashes with salted or keyed commitments, bind more of the routing catalog and tool-effect history, combine issuer receipts with receiver attestations and transparency logs, study how auditors use receipts in practice, and examine their role in conformity assessments. New commitments and corrected transport should be evaluated on their own revisions rather than credited with historical results.

9 Conclusion

Organizations need evidence with which responsible people can review and justify AI-assisted decisions. The public refund-dispute records make that need concrete. In Moffatt v. Air Canada, the customer's screenshot, not the airline's own records, established what its chatbot had said, and the airline did not produce the contract terms it relied on. This study addresses the run-level evidence needed within that broader organizational review process. It develops five design requirements and four principles and instantiates them in Lattice.

The historical controlled evaluation provides evidence about technical behavior: where rejection was required, single-receipt tamper attempts were rejected, retained outputs matched signed hashes, routing policies were not violated, and signing added milliseconds per run (Table 5). Separate hosted cohorts examine execution and answer quality. Later task, policy, and serialized-request commitments address identified gaps, but historical timings do not measure those extensions and request commitments do not prove remote consumption. Checking a stored output against its signed hash is verification of retained evidence, not re-execution of the model.

These results support a design for producing verifiable evidence during execution. They do not establish decision correctness, fairness, provider honesty, human authorization, or regulatory compliance. Self-attestation, unsalted hashes, optional verifier expectations, and incomplete external evidence remain substantive limits. Organizations must combine technical verification with retained business records, assigned review authority, and judgment about the decision itself. Whether this design helps managers investigate disputes or improves audit work remains a question for practitioner evaluation; no such benefit is claimed from the reported experiments.

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The personal-data evaluation uses the pii-masking-300k dataset by Ai4Privacy (Ai4Privacy, 2024). The `LICENSE.md` at the pinned revision grants the texts for academic research and non-commercial use when Ai4Privacy is acknowledged. Redistribution, uploading, sharing, and derivative works need a written licence from `licensing@ai4privacy.com`. Commercial use needs a prior discussion at that address; none was requested. This paper publishes aggregate scores only and does not redistribute the texts. One author's primary affiliation is Full Self Browsing, disclosed under competing interests. The Gretel synthetic finance dataset (Gretel.ai, 2024) is used under the Apache-2.0 license. The citation evaluation uses the ASQA dataset (Stelmakh et al., 2022) under the Apache-2.0 license.

Declarations

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Generative AI use During the preparation of this work, R. Neouchi used Claude (Claude Opus 5.5, Anthropic) to identify candidate journals, suggest a revision plan and manuscript structure, and identify literature for review. He reviewed its suggestions, and this use did not generate manuscript text. L. Turlapati used Claude Code, Cursor, Grok Bot, and OpenAI Codex as coding assistants. The author-reported models used across these tools were Claude Opus 5.5, Fable 5.1, GPT-6 Astra, GPT-6.1 Sol, Grok 4.7, and Composer 2.5. Documented Codex assistance also included evaluation implementation and analysis, report preparation, manuscript revision planning, drafting and editing evaluation-related manuscript text, editorial revision of the introduction, abstract, and conclusion, and preparation of the refund workflow diagrams and evidence tables. Codex also assisted source acquisition, protocol and rubric preparation, implementation, and analysis for the retrospective public-record evaluation. At L. Turlapati's direction, Claude Code (Claude Opus 5.5) drafted the narrative of Section 5 and the accompanying discussion of the public-record results, revised one sentence of the conclusion to match those records, re-verified the public-record sources, and audited the scoring rubric against them. The authors take responsibility for the content of this publication.

Competing interests Lakshman Turlapati is the author and maintainer of Lattice and is affiliated with Full Self Browsing, which publishes Lattice. Rabih Neouchi declares no competing interests.

Data and code availability The artifact and the evaluation harness, including the result files the tables are generated from, are available at `https://github.com/fullselfbrowsing/Lattice`. [TODO: Push the evaluation branch and archive a tagged release with a DOI (e.g., Zenodo) before submission.]

Author contributions [TODO: CRediT statement, e.g., Conceptualization: L.T., R.N.; Software: L.T.; Methodology: L.T., R.N.; Writing: L.T.; Supervision: R.N.]

Table 22: Per-run latency (ms) with a mocked OpenAI-shaped provider, 1000 trials after 100 warm-up. Overhead = signed run − unsigned run. Latency environment (`env-latency.json`): Node v24.22.1, Intel(R) Xeon(R) Processor x4, linux 6.12.94+, commit ae99fb5.

Config	plain p50	run p50	signed p50	signed p95	signed p99	verify p50
text-1KB	0.058	0.077	0.291	0.749	1.026	0.172
text-10KB	0.076	0.092	0.287	0.648	0.845	0.160
text-100KB	0.315	0.326	0.778	1.285	1.588	0.179
text-1000KB	3.529	3.494	5.592	7.672	9.616	0.221
image-100KB	0.062	0.083	0.422	0.825	1.213	0.167
image-1000KB	0.078	0.088	1.442	4.804	6.501	0.197
multimodal-5art	0.092	0.119	1.436	3.647	7.874	0.188
text-x1	0.050	0.070	0.285	0.516	0.756	0.166
text-x10	0.166	0.207	1.307	1.925	2.500	0.188
text-x50	0.535	0.746	5.888	7.531	13.557	0.217

Isolated: createReceipt p50 0.059 / p99 0.211; verifyReceipt p50 0.124 / p99 0.295; receiptCid p50 0.0500

Ethics approval Not applicable to the reported evaluation, which involved no human participants. [TODO: Required for the planned study in Sect. 7.12.]

A Reproducibility

The narrative quotes the harness through macros generated by `eval/scripts/06-report.py`. The latency and size results (Tables 22, 23, and 24; Figures 4 and 5) come from the original measurement run on Node v24.22.1, Intel(R) Xeon(R) Processor (4 cores, 15.6 GB), platform linux 6.12.94+, Lattice 1.6.0, commit `ae99fb5a3de3bb149542a0f61c55140661ef7600`, at 2026-10-26T18:11:44.232Z. The functional experiments (tamper, replay, policy, and verifier checks) do not depend on timing. They ran against the library source at commit `a8c1496661463333da8d1ac81fe97fa77e255b3b` on Node v24.14.1, Apple M4 Max (16 cores, 64 GB), platform darwin 25.6.0, at 2026-10-27T20:26:58.430Z, and the test counts below come from the same environment. The library tests passed 1782 of 1782 (0 failed). The command-line tests passed 196 of 196 (0 failed). Those counts document that the suite ran. They are not the argument for the design.

Provider calls were mocked. No API credential was read from the environment. Regenerate with `eval/run-all.sh` and `pnpm bench:receipts`. Both re-measure latency on the host that runs them. The repository benchmark, which is separate from the harness tamper suite, writes `paper/bench/results.json`. The live and comparison results of Sections 7.7 and 7.8 are generated from `eval/realworld/results` by `eval/realworld/scripts/05-charts.py` and `12-charts.py`.

Table 23: Receipt envelope size vs. input payload. Latency environment (`env-latency.json`): Node v24.22.1, Intel(R) Xeon(R) Processor x4, linux 6.12.94+, commit ae99fb5.

Config	payload B	artifacts	receipt B
text-1KB	1024	1	1183
text-10KB	10240	1	1183
text-100KB	102400	1	1183
text-1000KB	1024000	1	1183
image-100KB	102400	2	1271
image-1000KB	1024000	2	1271
multimodal-5art	618496	5	1539
text-x1	1024	1	1183
text-x10	10240	10	1987
text-x50	51200	50	5559

Table 24: Chained receipts (parentReceiptCid): size and verification cost by depth (30 trials). Latency environment (`env-latency.json`): Node v24.22.1, Intel(R) Xeon(R) Processor x4, linux 6.12.94+, commit ae99fb5.

Depth	total B	B/link	build p50 ms	verify-all p50 ms
1	1023	1023	0.166	0.267
2	2170	1085	0.270	0.426
4	4464	1116	0.525	0.949
8	9052	1132	1.070	1.728
16	18228	1139	1.934	3.135
32	36580	1143	3.839	6.357
64	73284	1145	7.272	12.149
128	146692	1146	14.322	23.619
256	293508	1147	29.848	47.278

Table 25: Offline replay over recorded fixtures, per adapter. Fixtures are verified with the public keyset written next to them. Functional environment (`env-functional.json`): Node v24.14.1, Apple M4 Max x16, darwin 25.6.0, commit a8c1496.

Adapter	runs	verified	hash match	live rerun match	altered detected
openai	50	50	50	50	50
openai-compatible	50	50	50	50	50
anthropic	50	50	50	50	50
gemini	50	50	50	50	50
xai	50	50	50	50	50
openrouter	50	50	50	50	50
lm-studio	50	50	50	50	50

Table 26: Policy checks on a synthetic labeled set (100 samples per label). The original set was written alongside the detectors. The added set holds PII in support and CRM prose, tables, and CSV rows, more card networks, and uncued numbers that are not PII. Rows with only the original negatives leave the added negatives out. The mustCite cases are the original set, default-behavior cases, and opt-in prose checks. The prose-by-default row scores the original mustCite cases against labels that expect prose to be read by default; inline-text-only and hallucinated-cite-string differ, because the default reads prose only on opt-in. The set is a regression suite, not a prevalence-weighted estimate. Functional environment (`env-functional.json`): Node v24.14.1, Apple M4 Max x16, darwin 25.6.0, commit a8c1496.

Set	n	precision	recall	FP	FN
noPII, in-scope PII types, all negatives	5500	1	0.9429	0	200
noPII, in-scope PII types, original negatives only	4500	1	0.9429	0	200
noPII, all PII types, all negatives	6100	1	0.8083	0	786
noPII, all PII types, original negatives only	5100	1	0.8083	0	786
noPII, out-of-scope types, all negatives	2600	1	0.0233	0	586
noPII, original labels only (in-scope)	2400	1	1	0	0
noPII, added labels only (in-scope)	3100	1	0.9048	0	200
added uncued numeric negatives	1000	0 flagged			
mustCite agreement, all cases	36	36/36 agree with label			
mustCite, default behavior	22	22/22			
mustCite, opt-in prose checks	14	14/14			
mustCite, original cases	15	15/15			
mustCite, original cases, prose-by-default labels	15	13/15 (2 labels differ)			
routing determinism	20000	identical 1.0000, permuted same 1.000, violations 0			

Table 27: Tamper suite over 70 receipts from mocked `ai.run` calls across the seven adapters: rejections per mutation and the error kind `verifyReceipt` returned. Rows in the control-accept and replay classes are expected to verify (there is no replay cache). Last block: 50 linked chains of 8 receipts, rejection rate by per-receipt `verifyReceipt` / by the library's `verifyChain` with no `expectedCount` (except the expectedCount row). Accepted chains list the advisory codes `verifyChain` returned. Functional environment (`env-functional.json`): Node v24.14.1, Apple M4 Max x16, darwin 25.6.0, commit a8c1496.

Class	Mutation	Rejected	Kind
body-edit-keep-sig	edit-outputHash	70/70	signature-invalid:70
body-edit-jcs-keep-sig	edit-outputHash	70/70	signature-invalid:70
body-edit-attacker-sig-legit-kid	edit-outputHash	70/70	signature-invalid:70
body-edit-keep-sig	edit-inputHashes	70/70	signature-invalid:70
body-edit-jcs-keep-sig	edit-inputHashes	70/70	signature-invalid:70
body-edit-attacker-sig-legit-kid	edit-inputHashes	70/70	signature-invalid:70
body-edit-keep-sig	edit-route-provider	70/70	signature-invalid:70
body-edit-jcs-keep-sig	edit-route-provider	70/70	signature-invalid:70
body-edit-attacker-sig-legit-kid	edit-route-provider	70/70	signature-invalid:70
body-edit-keep-sig	edit-model	70/70	signature-invalid:70
body-edit-jcs-keep-sig	edit-model	70/70	signature-invalid:70
body-edit-attacker-sig-legit-kid	edit-model	70/70	signature-invalid:70
body-edit-keep-sig	edit-usage-cost	70/70	signature-invalid:70
body-edit-jcs-keep-sig	edit-usage-cost	70/70	signature-invalid:70
body-edit-attacker-sig-legit-kid	edit-usage-cost	70/70	signature-invalid:70
body-edit-keep-sig	edit-verdict	70/70	signature-invalid:70
body-edit-jcs-keep-sig	edit-verdict	70/70	signature-invalid:70
body-edit-attacker-sig-legit-kid	edit-verdict	70/70	signature-invalid:70
body-edit-keep-sig	edit-issuedAt	70/70	signature-invalid:70
body-edit-jcs-keep-sig	edit-issuedAt	70/70	signature-invalid:70
body-edit-attacker-sig-legit-kid	edit-issuedAt	70/70	signature-invalid:70
body-edit-keep-sig	edit-runId	70/70	signature-invalid:70
body-edit-jcs-keep-sig	edit-runId	70/70	signature-invalid:70
body-edit-attacker-sig-legit-kid	edit-runId	70/70	signature-invalid:70
body-edit-keep-sig	drop-redactions	70/70	version-mismatch:70
body-edit-jcs-keep-sig	drop-redactions	70/70	version-mismatch:70
body-edit-attacker-sig-legit-kid	drop-redactions	70/70	version-mismatch:70
body-edit-keep-sig	edit-redactionPolicy	70/70	signature-invalid:70
body-edit-jcs-keep-sig	edit-redactionPolicy	70/70	signature-invalid:70
body-edit-attacker-sig-legit-kid	edit-redactionPolicy	70/70	signature-invalid:70
body-edit-keep-sig	add-field	70/70	canonicalization-mismatch:70
body-edit-jcs-keep-sig	add-field	70/70	signature-invalid:70
body-edit-attacker-sig-legit-kid	add-field	70/70	signature-invalid:70
body-edit-keep-sig	unicode-nfd	70/70	signature-invalid:70
body-edit-jcs-keep-sig	unicode-nfd	70/70	signature-invalid:70
body-edit-attacker-sig-legit-kid	unicode-nfd	70/70	signature-invalid:70
encoding	key-reorder	70/70	canonicalization-mismatch:70
encoding	whitespace	70/70	canonicalization-mismatch:70
signature	sig-bitflip	70/70	signature-invalid:70

Table 27, continued

Class	Mutation	Rejected	Kind
signature	sig-swap-same-key	70/70	signature-invalid:70
signature	sig-truncate	70/70	signature-invalid:70
signature	sig-empty-list	70/70	envelope-malformed:70
signature	payload-truncate	70/70	envelope-malformed:70
signature	payloadType-change	70/70	envelope-malformed:70
key	keyid-to-other-trusted	70/70	signature-invalid:70
key	keyid-unknown	70/70	key-not-found:70
key	forge-attacker-key-own-kid	70/70	key-not-found:70
key	resign-revoked-key	70/70	key-revoked:70
key	resign-other-key-body-kid-mismatch	70/70	signature-invalid:70
downgrade	version-v1-resigned-valid-key	70/70	schema-version-too-low:70
downgrade	version-absent-resigned-valid-key	70/70	schema-version-too-low:70
control-accept	untouched	0/70	accepted(active):70
control-accept	retired-key-legit-signature	0/70	accepted(retired):70
control-accept	version-v1.1-resigned-valid-key (key holder)	0/70	accepted(active):70
replay	same-receipt-presented-again (no nonce/audience)	0/70	accepted(active):70
Chain attacks: rejection rate, per-receipt verify / verifyChain			
chain	swap-adjacent-links	0.00 / 0.00	accepted: tail-truncation-unchecked, audience-absent, expiry-absent, nonce-absent
chain	drop-middle-link	0.00 / 1.00	rejected: orphan-link
chain	splice-foreign-link	0.00 / 1.00	rejected: orphan-link
chain	truncate-tail	0.00 / 0.00	accepted: tail-truncation-unchecked, audience-absent, expiry-absent, nonce-absent
chain	truncate-tail-expectedCount	0.00 / 1.00	rejected: count-mismatch
chain	replace-root	0.00 / 1.00	rejected: orphan-link

Table 28: Verifier opt-ins, structured advisories, chain closure, crew receipts, and signer-failure modes: 71 of 71 checks pass. Result gives the error kind, advisory, or exception the check expected. The two base-1.6.0 rows run the released 1.6.0 verifier on a chain minted by the library at the functional commit. Functional environment (`env-functional.json`): Node v24.14.1, Apple M4 Max x16, darwin 25.6.0, commit a8c1496.

Check	Result
plain-verifies	pass
advisories-on-unbound-receipt	pass: 3 info advisories
advisory-shape-structured	pass: {code, severity, message}
current-version-no-version-advisory	pass
bound-accepts-matching-options	pass
audience-mismatch	pass: audience-mismatch
nonce-mismatch	pass: nonce-mismatch
expired	pass: receipt-expired

Table 28, continued

Check	Result
minVersion-v1.4-accepts-current	pass
bound-unchecked-audience-warns	pass: warn:audience-unchecked
bound-unchecked-nonce-warns	pass: warn:nonce-unchecked
nonce-value-never-echoed	pass: not echoed
expiresAt-set-no-expiry-advisory	pass
bound-and-checked-no-advisories	pass: 0 advisories
maxAgeMs-silences-expiry-absent	pass
minVersion-floor-rejects-v1.1	pass: schema-version-too-low
v1.1-still-verifies-without-minVersion	pass
version-below-current-info-effective-floor	pass: info:version-below-current
version-below-current-kept-at-baseline-minVersion	pass: info:version-below-current
truncate-without-expectedCount	pass: accepted
truncate-with-expectedCount	pass: count-mismatch
expectedCount-silences-tail-advisory	pass: accepted
requireClosure-rejects-unclosed-chain	pass: count-mismatch
closed-chain-accepts	pass
closed-chain-no-tail-advisory	pass
closed-chain-dropped-leaf	pass: count-mismatch
closed-chain-dropped-closer	pass: count-mismatch
requireClosure-accepts-closed-chain	pass
partial-subset-accepts-with-closure-partial-warn	pass: warn:closure-partial
partial-full-set-no-closure-partial	pass: accepted
subset-without-partial-rejected	pass: count-mismatch
partial-requires-closer	pass: count-mismatch
partial-rejects-unlisted-member	pass: count-mismatch
createReceipt-rejects-malformed-member-cid	pass: TypeError
createReceipt-rejects-non-array-member-list	pass: TypeError
createReceipt-rejects-duplicate-member-cid	pass: TypeError
createReceipt-rejects-unsorted-member-list	pass: TypeError
createReceipt-rejects-unknown-closure-mode	pass: TypeError
createReceipt-rejects-closure-on-non-root	pass: TypeError
malformed-chainClosure-mode-verifyReceipt	pass: version-mismatch
malformed-chainMemberCids-string-verifyReceipt	pass: version-mismatch
malformed-chainMemberCids-non-string-verifyReceipt	pass: version-mismatch
malformed-root-closure-verifyChain-link-invalid	pass: link-invalid(version-mismatch)
malformed-closer-verifyChain-link-invalid	pass: link-invalid(version-mismatch)
malformed-closer-partial-link-invalid	pass: link-invalid(version-mismatch)
duplicate-member-cid-verifyChain	pass: count-mismatch
non-root-closure-receipt-verifies-alone	pass
non-root-closure-verifyChain-rejected	pass: count-mismatch
non-root-closure-partial-rejected	pass: count-mismatch
root-only-closure-unclosed-root	pass: count-mismatch
crew-receipts-n-plus-2	pass: 5 receipts for 3 agents
crew-start-declares-closure-closer-lists-members	pass: closer lists 4
crew-chain-verifies-requireClosure	pass

Table 28, continued

Check	Result
crew-closer-verdict-success	pass: success
crew-dropped-leaf-count-mismatch	pass: count-mismatch
crew-dropped-closer-count-mismatch	pass: count-mismatch
crew-partial-disclosure-warns	pass: warn:closure-partial
crew-budget-exceeded-closer-execution-failed	pass: crew-budget-exceeded / execution-failed
crew-child-mint-failure-default-receiptError	pass: signer-failed
crew-child-mint-failure-flag-receiptError	pass: signer-failed
crew-child-mint-failure-throw-ReceiptSigningError	pass: ReceiptSigningError
crew-child-mint-throw-stops-crew	pass: 2 provider calls
crew-closer-mint-failure-flagged-chain-unverifiable	pass: count-mismatch
crew-start-failure-default-ReceiptSigningError	pass: ReceiptSigningError
crew-start-failure-flag-ReceiptSigningError	pass: ReceiptSigningError
crew-start-failure-throw-ReceiptSigningError	pass: ReceiptSigningError
base-1.6.0-verifier-accepts-new-crew-chain	pass: latticeVersion 1.6.0
base-1.6.0-verifier-does-not-enforce-closure (documented)	pass: accepted
signer-flag-default	pass: signer-failed
signer-throw	pass: ReceiptSigningError
no-signer-no-receipt	pass

B Earlier synthetic workflow controls

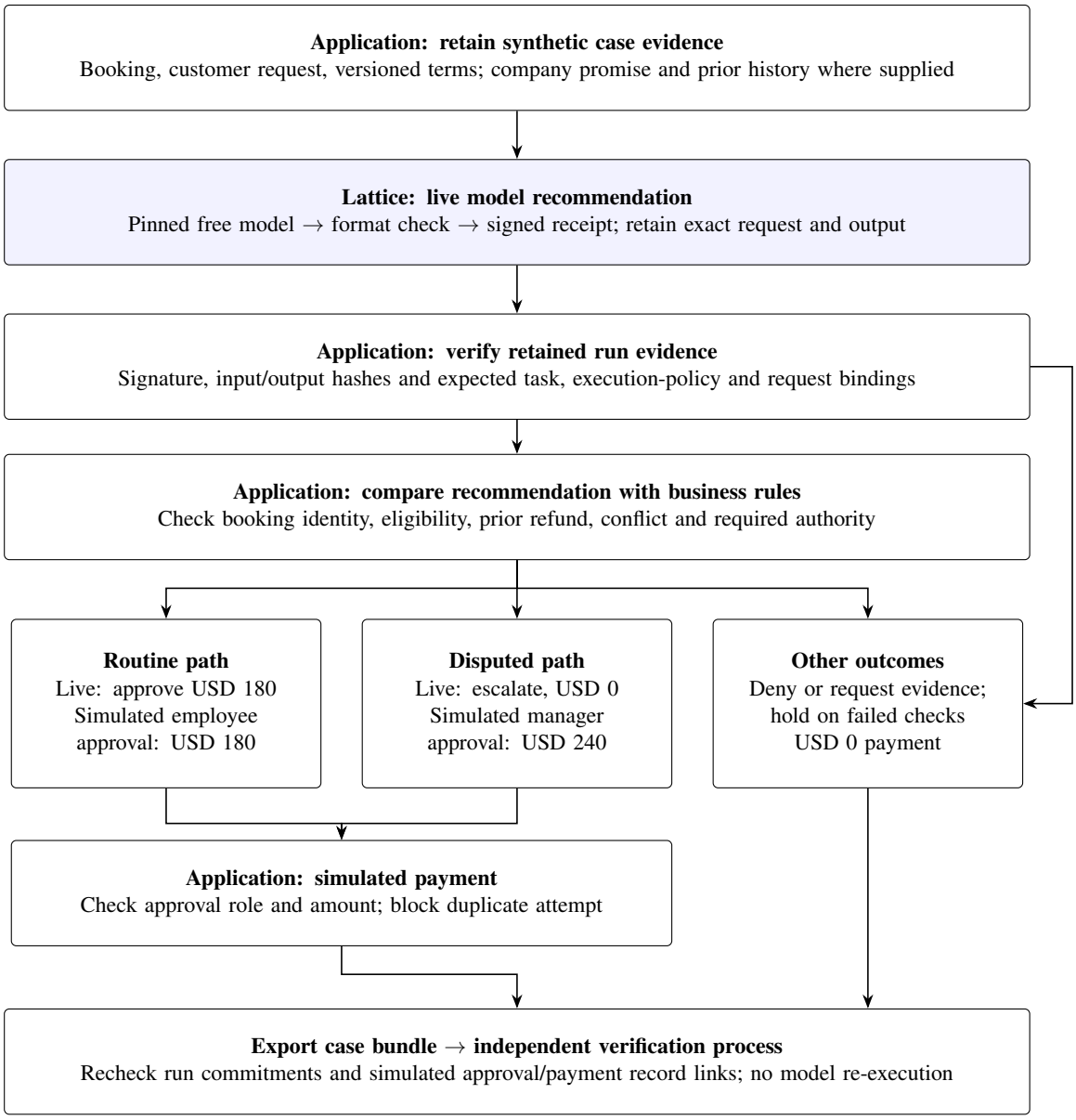


Figure 19: Earlier synthetic workflow, retained as a technical control. Recommendations are live model outputs; all staff approvals and payments are simulated. The disputed case's earlier denial is scripted backstory. Failed-check holds were exercised in separate controls; none occurred in the live cohort. Verification checks retained evidence, not actual employee identity or settlement (Table 29); measured outcomes appear in Table 30. Diagram prepared with OpenAI Codex; drafting model/version unconfirmed, pending author disclosure review.

Table 29: Evidence and responsibility in the synthetic refund workflow

Step / responsible role	Retained evidence	Implemented check	Remaining boundary
Intake / application operator	Booking, request, versioned terms; company promise and prior history for the dispute	Compare exact artifact values with signed input hashes and expected fixture inventory	No proof of source authenticity, truthful history or complete real-world capture
Run / Lattice operator	Task, execution policy, serialized request, output and issuer-signed receipt	Signature plus expected task, execution-policy and request bindings	Business terms are separately hashed inputs; no proof of remote request consumption or issuer honesty
Recommendation / application	Action, amount, authority, conflict flag, rule IDs and source list	Format invariant; separate application comparison of decision fields with business rules	Format does not enforce required-source coverage; citation entailment and fairness are untested
Review / employee or manager fixture	Approval role, amount, reason, case ID and recommendation hash; denial-review event	Role/amount guards and linkage to the recommendation	Simulated approval is not authentication of a person or proof of delegated authority
Payment / payment-service fixture	Approval link, amount, outcome and duplicate-attempt result	Approval/amount checks; in-memory duplicate guard; record-link checks	No real settlement, durable transaction concurrency or independently authenticated payment event
Audit / separate verifier process	Trusted public key, exported run evidence and simulated application records	Recompute commitments; check expected inventory and approval/payment links	Requires trusted expectations; retained-output verification is not re-execution, correctness or regulatory compliance

Table 30: Live free-model results for the synthetic refund demonstration (separate from historical cohorts)

Measure	Nemotron 3.5 Super VL	Dots 3 Note Preview
Completed evaluation runs	16/16	16/16
Decision and rule checks	16/16	16/16
Full rubric, including required references	14/16	14/16
Verified run-evidence packages	16/16	16/16

Cohort. 8 synthetic scenarios × 2 models × 2 repetitions = 32 planned runs; 2 development calls excluded. Measured revision: `f8a6914`. Temperature 0, output cap 4096 tokens; reasoning settings omitted (provider defaults). Pinned OpenRouter IDs: `nvidia/nemotron-3.5-super-vl:free` and `dots-studio/dots-3-note-preview:free`. No retries or post-freeze prompt repair.

Scoring. Decision checks cover action, immediate amount, authority, conflict flag and decisive-rule inclusion. Full-rubric passes: 28/32. All 4 failures omitted the customer-request reference: Nemotron 3.5 Super VL, already refunded (1 run); Nemotron 3.5 Super VL, customer instruction (1 run); Dots 3 Note Preview, disputed promise (2 runs). The runtime format contract accepted these outputs; required-source coverage was scored separately. Explanation entailment was not systematically graded.

Verification and controls. 32/32 exported run-evidence packages verified in a fresh process (signature, input/output content and expected task, execution-policy and serialized-request bindings). 18/18 scoped integrity/application checks passed on one routine bundle, including intact controls; these are not 18 distinct cryptographic attacks. A separate scripted offline wrong-but-signed control had valid evidence, failed decision scoring and was held by application rules.

Cost and scope. All 34 inference requests reported USD 0 cost; added account usage was USD 0. Recommendations used live models; approvals and payments were simulated. The cohort does not measure operational accuracy, human review outcomes or superiority over secured logging. Verification is of retained records, not model re-execution.

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